Tight PAC-Bayes Bounds for Deep Neural Networks

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1 Introduction

Open Problem (solveall.org). Determine whether there exists a PAC-Bayes pipeline (choice of prior/posterior family, optimization procedure, and certificate computation) that simultaneously satisfies all three goals below for modern deep-learning practice:

1. **Non-vacuity at realistic scale:** on standard trained high-capacity models (including ImageNet-scale settings), certify 0–1 upper bounds $B(S, \delta) < 1$ with confidence $1 - \delta$.
2. **Polynomial-time certifiability:** compute the bound (including numerical certification of all terms) in polynomial time.
3. **Predictive tightness across choices:** over a family of training pipelines or hyperparameters Λ, the values $\{B_\lambda(S, \delta) : \lambda \in \Lambda\}$ should strongly track true test risks (for example, high rank correlation), not merely be non-vacuous.

Prior work achieved these criteria in isolation or pairwise, but no unified framework was known to satisfy all three simultaneously with rigorous theoretical guarantees and empirical verification at practical scale.

Our answer is yes in the following precise conditional sense. We give a rigorous sample-split PAC-Bayes framework for Gibbs predictors, instantiate it on an explicit finite ImageNet family, and discharge the audit-side certification step via a two-sided empirical-Bernstein audit run (Audibert–Munos–Szepesvári, 2009) (Section 12, Appendix A). The framework uses:

1. a **PAC-Bayes-kl inequality for measurable posterior kernels** with Maurer's constant $C(n, \delta) = \ln(2\sqrt{n}/\delta)$,
2. **kl-inverse certificate computation** via certified bisection with explicit numerical error tracking,
3. **Scope of the present theorem package**: sample splitting for Gibbs predictors together with an independent audit set for predictive-tightness certification.

Validity is universal over all admissible prior-posterior pairs with finite KL inside this sample-split setup; polynomial-time computability (Theorem 7.2) additionally requires that one can draw $\theta \sim Q$ and compute a certified upper bound $\mathrm{KL}_{ub}(Q) \geq \mathrm{KL}(Q\|P)$ in polynomial time from the representation of (P, Q).

Scope caveat. The present package resolves the three criteria within the sample-split Gibbs-predictor setting. A stronger program would replace the sample-split with a full-data PAC-Bayes regime and add an explicit transfer theorem from Gibbs risk to the corresponding deterministic predictor; this is separate from the theorem chain established here and is discussed only briefly in the Conclusion.

2 Proof Construction

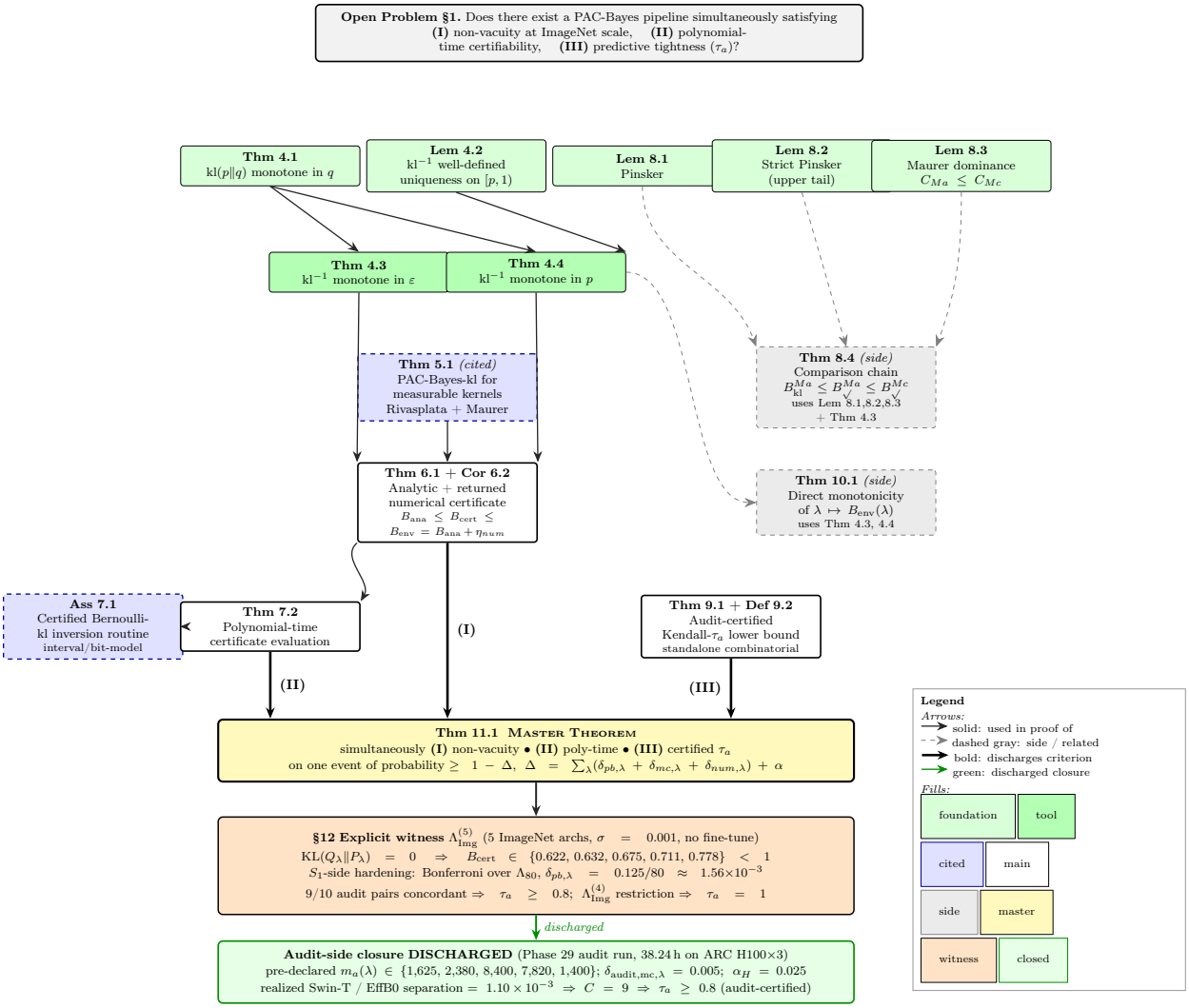

Figure 1: Proof construction: how each result is built on top of the previous. Solid black arrows mark logical dependency; dashed gray arrows mark side results (Comparison theorem, Direct monotonicity). The three bold pillar arrows labeled **(I)** non-vacuity, **(II)** polynomial-time, **(III)** certified Kendall-τ_a discharge the three criteria of the Open Problem into the Master Theorem (Thm 11.1). The green arrow records that the audit-side Monte-Carlo closure is discharged on $\Lambda^{(5)}_{\mathrm{Img}}$ by the Phase 29 audit run ($C = 9$, $\tau_a \ge 0.8$).

3 Setup and Notation

Let $\mathcal{X}$ be an input space, $\mathcal{Y} = \{1, \dots, C\}$ a finite label set, and $\mathcal{D}$ an unknown distribution on $\mathcal{X} \times \mathcal{Y}$. A training sample

$$S = ((x_1, y_1), \dots, (x_n, y_n)) \sim \mathcal{D}^n$$

is split into disjoint subsamples $S = S_0 \sqcup S_1$ with $|S_1| = n_1$. The split is taken to be a fixed deterministic index split, or a random split independent of the data and of the audit sample, conditioned on its realization. The prior is constructed from S_0 and is independent of S_1; the posterior may depend measurably on both S_0 and S_1. The sample S_1 is the sample that appears in the PAC-Bayes-kl certificate. For predictive-tightness certification we additionally use an audit sample $T = ((x'_1, y'_1), \dots, (x'_{n_2}, y'_{n_2})) \sim \mathcal{D}^{n_2}$, independent of (S_0, S_1).

All certificate-side Monte Carlo randomness (the i.i.d. draws $\theta_{\lambda,j} \sim Q_\lambda$ used to form $\hat{L}_{S_1}^{(m)}$, and any other randomness appearing in the computation of B_{cert}) is generated independently of the audit sample T conditional on (S_0, S_1). This convention discharges the "U independent of T conditional on (S_0, S_1)" hypothesis of Theorem 9.1 when applied in the Master Theorem.

Let $h_\theta : \mathcal{X} \to \mathcal{Y}$ denote the predictor parametrized by $\theta \in \Theta$, where $\Theta \subseteq \mathbb{R}^d$ is a d-dimensional parameter space. When applied to a finite family Λ of architecture configurations, each pair (P_λ, Q_λ) may be defined on its own measurable space $(\Theta_\lambda, \Sigma_{\Theta_\lambda})$ with $\Theta_\lambda \subseteq \mathbb{R}^{d_\lambda}$ and architecture-specific dimension d_λ; equivalently, take $\Theta := \bigsqcup_{\lambda\in\Lambda}\{\lambda\} \times \Theta_\lambda$ as a common disjoint-union space with each P_λ, Q_λ supported on its own component, and apply the PAC-Bayes-kl inequality of Theorem 5.1 per λ.

Throughout, we treat each predictor h_θ as the deterministic finite-precision program implementing it, so that empirical losses $\hat{L}_{S_1}(h_\theta)$ are evaluated exactly in the bit model conditional on θ. If empirical losses are not evaluated exactly, an upper bound $\gamma_{\text{eval}} \geq 0$ on the total per-sample evaluation error must be added to the certificate input, i.e., one replaces p_{cert} in Theorem 6.1 by $\min\{1, \hat{L}_{S_1}^{(m)}(Q) + \varepsilon_{mc} + \gamma_{\text{eval}}\}$, or else propagates γ_{eval} through a certified Lipschitz/monotonicity calculation into the returned endpoint B_{cert}. The kl-inversion tolerance η_{num} governs only the bracket width $B_{\text{cert}} - B_{\text{ana}}$ at the inversion step, not forward-pass approximation error.

For the empirical witness of Section 12, the loss ℓ is the bounded 0-1 indicator, so each per-sample evaluation $\ell(h_\theta(x_i), y_i) \in \{0, 1\}$ is an exact program output. Accumulating these indicators as exact integer counts, the certificate-side MC mean is the exact rational $\hat{L}_{S_1}^{(m)}(Q_\lambda) = K_\lambda^{\text{cert}}/(n_1 m)$ for some integer $K_\lambda^{\text{cert}} \in \{0, 1, \ldots, n_1 m\}$, where $n_1 = |S_1|$ and $m = m(\lambda)$, and is propagated into the certified kl bisection either as an exact rational or as an outward-rounded floating-point interval enclosure. Symmetrically, the audit-side MC mean is the exact rational $\hat{L}_T^{(m_a)}(Q_\lambda) = K_\lambda^{\text{audit}}/(n_2 m_a)$ for some integer $K_\lambda^{\text{audit}} \in \{0, 1, \ldots, n_2 m_a\}$, where $n_2 = |T|$ and $m_a = m_a(\lambda)$, and is propagated into the audit-side empirical-Bernstein interval calculation. In both cases the forward-pass-evaluation error $\gamma_{\text{eval}} = 0$ for the formal certificate and any rounding in the displayed decimal summaries is not used in the certificate computation; no additional padding is needed. We consider the Gibbs predictor induced by posterior Q on Θ with bounded 0-1 loss $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, 1]$.

Key quantities:

$$L_{\mathcal{D}}(Q) := \mathbb{E}_{\theta\sim Q}[\mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h_\theta(x), y)]], \tag{1}$$

$$\hat{L}_S(Q) := \mathbb{E}_{\theta\sim Q}\left[\frac{1}{n}\sum_{i=1}^{n} \ell(h_\theta(x_i), y_i)\right]. \tag{2}$$

When evaluated on the evaluation subset S_1 of size n_1, we write $\hat{L}_{S_1}(Q)$ with the sum taken over S_1 and denominator n_1. For a single parameter θ, we write

$$\hat{L}_{S_1}(h_\theta) := \frac{1}{n_1}\sum_{i\in S_1} \ell(h_\theta(x_i), y_i).$$

When evaluated on the audit subset T of size n_2, we write $\hat{L}_T(Q)$ for the corresponding Gibbs empirical loss on T.

Bernoulli KL and its upper inverse. For $p, q \in [0, 1]$, we use the usual lower-semicontinuous extension of the Bernoulli relative entropy

$$\mathrm{kl}(p\|q) = p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}$$

to the boundary of $[0, 1]^2$. Explicitly,

$$\mathrm{kl}(0\|q) = \ln \tfrac{1}{1-q} \quad (0 \leq q < 1), \qquad \mathrm{kl}(1\|q) = \ln \tfrac{1}{q} \quad (0 < q \leq 1),$$

$$\mathrm{kl}(p\|0) = +\infty \quad (p > 0), \qquad \mathrm{kl}(p\|1) = +\infty \quad (p < 1), \qquad \mathrm{kl}(0\|0) = \mathrm{kl}(1\|1) = 0.$$

Terms of the form $0 \ln 0$ are interpreted by continuity as 0. For $p \in [0, 1]$ and finite $\varepsilon \geq 0$, define

$$\mathrm{kl}_+^{-1}(p, \varepsilon) := \max\{q \in [p, 1] : \mathrm{kl}(p\|q) \leq \varepsilon\}.$$

The maximum is well-defined: see Lemma 4.2 below.

Certificate parameters. The certified bound (Theorem 6.1) requires several error-tolerance and approximation parameters beyond the standard PAC-Bayes setup:

- $\mathrm{KL}_{ub}(Q) \geq \mathrm{KL}(Q\|P)$: a certified upper bound on the KL divergence, assumed polynomial-time computable from the representation of (P, Q).
- $\hat{L}_{S_1}^{(m)}(Q) := \frac{1}{m}\sum_{j=1}^{m} \hat{L}_{S_1}(h_{\theta_j})$ with $\theta_j \overset{\text{iid}}{\sim} Q$: Monte Carlo estimate of $\hat{L}_{S_1}(Q)$.
- ε_{mc}: Monte Carlo tolerance. Unless $\hat{L}_{S_1}(Q)$ is computed exactly, we assume $\varepsilon_{mc} > 0$, $\delta_{mc} \in (0,1)$, and $m \geq \lceil \ln(2/\delta_{mc})/(2\varepsilon_{mc}^2) \rceil$. Then Hoeffding's inequality gives $|\hat{L}_{S_1}^{(m)}(Q) - \hat{L}_{S_1}(Q)| \leq \varepsilon_{mc}$ with probability $\geq 1-\delta_{mc}$, conditional on (S_0, S_1). In the exact-computation case we set $\varepsilon_{mc} = 0$, $\delta_{mc} = 0$, and interpret $\hat{L}_{S_1}^{(m)}(Q) := \hat{L}_{S_1}(Q)$.
- $\eta_{num} \geq 0$: bisection bracket tolerance for numerical computation of kl_+^{-1}.
- $B_{\mathrm{ana}}(Q) := \mathrm{kl}_+^{-1}(p_{\mathrm{cert}}, \varepsilon_{pb}^{\mathrm{cert}}(Q))$: the exact analytic kl-inverse certificate, proved valid in Theorem 6.1.
- $B_{\mathrm{cert}}(Q)$: the returned certified numerical upper endpoint produced by a certified inversion routine (Assumption 7.1). It is required to satisfy $B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{ana}}(Q) + \eta_{num}$. This is the computable certificate value used throughout for non-vacuity and ranking.
- $B_{\mathrm{env}}(Q) := B_{\mathrm{ana}}(Q) + \eta_{num}$: the formal analytic-plus-tolerance envelope, a proof device used only where the argument explicitly concerns this slack expression (see Theorem 10.1). The three objects satisfy $B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{env}}(Q)$.
- $C(n, \delta) := \ln(2\sqrt{n}/\delta)$: the Maurer constant [4].
- $\delta_{pb}, \delta_{mc}, \delta_{num}$: PAC-Bayes, Monte Carlo, and numerical confidence budgets. The analytic certificate theorem uses only δ_{pb} and δ_{mc}; a separate numerical budget δ_{num} is needed only if the inversion routine itself is randomized. For the deterministic bisection used here, $\delta_{num} = 0$ suffices.

4 Monotonicity Properties of kl_+^{-1}

These monotonicity lemmas are foundational to the proofs of the certificate correctness, comparison, and direct monotonicity theorems.

Theorem 4.1 (Monotonicity of $\mathrm{kl}(p\|q)$ in q)**.** *For fixed $p \in [0,1)$, the map $q \mapsto \mathrm{kl}(p\|q)$ is continuous and strictly increasing on $[p,1)$, with $\mathrm{kl}(p\|p) = 0$ and $\lim_{q\uparrow 1} \mathrm{kl}(p\|q) = +\infty$.*

Proof. For $p \in (0,1)$ and $q \in (p,1)$:

$$\frac{\partial}{\partial q}\mathrm{kl}(p\|q) = -\frac{p}{q} + \frac{1-p}{1-q}.$$

Since $q > p$, we have $p/q < 1$ and $(1-p)/(1-q) > 1$, so the derivative is positive. Hence $q \mapsto \mathrm{kl}(p\|q)$ is strictly increasing on $(p,1)$. Continuity is clear from the definition. For the boundary: $\mathrm{kl}(p\|p) = p\ln 1 + (1-p)\ln 1 = 0$, and $\lim_{q\uparrow 1}(1-p)\ln\frac{1-p}{1-q} = +\infty$ since $p < 1$. The case $p = 0$ gives $\mathrm{kl}(0\|q) = -\ln(1-q)$, which is strictly increasing on $[0,1)$ with limit $+\infty$. □

Lemma 4.2 (Well-definedness of kl_+^{-1})**.** *For every $p \in [0,1]$ and every finite $\varepsilon \geq 0$, the set $\{q \in [p,1] : \mathrm{kl}(p\|q) \leq \varepsilon\}$ is nonempty and has a maximum. If $p < 1$, then $\mathrm{kl}_+^{-1}(p,\varepsilon) \in [p,1)$ and is the unique $q \in [p,1)$ satisfying $\mathrm{kl}(p\|q) = \varepsilon$. If $p = 1$, then $\mathrm{kl}_+^{-1}(1,\varepsilon) = 1$.*

Proof. The set is nonempty because $q = p$ is feasible ($\mathrm{kl}(p\|p) = 0$). If $p = 1$, the feasible set is contained in $[1,1] = \{1\}$, so the maximum is 1.

Now suppose $p < 1$. By Theorem 4.1, the map $q \mapsto \mathrm{kl}(p\|q)$ is continuous and strictly increasing on $[p,1)$, equals 0 at $q = p$, and tends to $+\infty$ as $q \uparrow 1$. Hence for every finite $\varepsilon \geq 0$ there is a unique $q_\varepsilon \in [p,1)$ with $\mathrm{kl}(p\|q_\varepsilon) = \varepsilon$. The feasible set is exactly $[p, q_\varepsilon]$, so its maximum is q_ε. □

Theorem 4.3 (Monotonicity of kl_+^{-1} in ε)**.** *For fixed $p \in [0,1]$, if $0 \le \varepsilon \le \varepsilon'$, then $\mathrm{kl}_+^{-1}(p,\varepsilon) \le \mathrm{kl}_+^{-1}(p,\varepsilon')$. The inequality is strict when $p < 1$ and $\varepsilon < \varepsilon'$.*

Proof. Weak monotonicity. The feasible set $\{q \in [p,1] : \mathrm{kl}(p\|q) \le \varepsilon\}$ is contained in $\{q \in [p,1] : \mathrm{kl}(p\|q) \le \varepsilon'\}$, so the supremum over the larger set is at least as large.

Strict monotonicity. Suppose $p < 1$ and $\varepsilon < \varepsilon'$. Let $q^* = \mathrm{kl}_+^{-1}(p,\varepsilon)$. By Theorem 4.1, $\mathrm{kl}(p\|\cdot)$ is continuous and strictly increasing on $[p,1)$ with $\mathrm{kl}(p\|p) = 0$ and $\lim_{q\uparrow 1} \mathrm{kl}(p\|q) = +\infty$, so for finite ε the supremum in the definition of kl_+^{-1} is attained at the unique $q^* \in [p,1)$ satisfying $\mathrm{kl}(p\|q^*) = \varepsilon$. Thus $\mathrm{kl}(p\|q^*) = \varepsilon < \varepsilon'$, and there exists $q' > q^*$ with $\mathrm{kl}(p\|q') = \varepsilon'$ (by the intermediate value theorem). Hence $\mathrm{kl}_+^{-1}(p,\varepsilon') \ge q' > q^* = \mathrm{kl}_+^{-1}(p,\varepsilon)$. □

Theorem 4.4 (Monotonicity of kl_+^{-1} in p)**.** *For fixed $\varepsilon \ge 0$, if $0 \le p \le p' \le 1$, then $\mathrm{kl}_+^{-1}(p,\varepsilon) \le \mathrm{kl}_+^{-1}(p',\varepsilon)$. The inequality is strict when $\varepsilon > 0$ and $p < p'$.*

Proof. Weak monotonicity. Let $q^* = \mathrm{kl}_+^{-1}(p,\varepsilon)$. We consider two cases.

Case 1: $q^* < p'$. Then $q^* \le p' \le \mathrm{kl}_+^{-1}(p',\varepsilon)$, since $\mathrm{kl}_+^{-1}(p',\varepsilon) \ge p'$ by definition.

Case 2: $q^* \ge p'$. For $q \ge p' \ge p$, we claim $\mathrm{kl}(p'\|q) \le \mathrm{kl}(p\|q)$. Indeed, for $p \in (0,q)$ the partial derivative $\frac{\partial}{\partial p}\mathrm{kl}(p\|q) = \ln\frac{p(1-q)}{q(1-p)}$ is strictly negative (since $p(1-q) < q(1-p)$ iff $p < q$), so $p \mapsto \mathrm{kl}(p\|q)$ is strictly decreasing on $(0,q)$. At $p = 0$ the conclusion $\mathrm{kl}(0\|q) \ge \mathrm{kl}(p'\|q)$ follows from continuity of $p \mapsto \mathrm{kl}(p\|q)$ on $[0,q)$ combined with this strict decrease: letting $p \downarrow 0$ in $\mathrm{kl}(p\|q) \ge \mathrm{kl}(p'\|q)$ for $p \in (0,p')$. Since $p \le p' \le q^*$, we have $\mathrm{kl}(p'\|q^*) \le \mathrm{kl}(p\|q^*) \le \varepsilon$. So q^* is feasible for (p',ε), giving $\mathrm{kl}_+^{-1}(p',\varepsilon) \ge q^*$.

Strict monotonicity. Suppose $\varepsilon > 0$ and $p < p'$. If $p' = 1$, then $\mathrm{kl}_+^{-1}(1,\varepsilon) = 1$ by Lemma 4.2, while $\mathrm{kl}_+^{-1}(p,\varepsilon) < 1$ (since $p < 1$ and Lemma 4.2 gives $\mathrm{kl}_+^{-1}(p,\varepsilon) \in [p,1)$), so the inequality is strict. Now suppose $p' < 1$. In Case 1, $\mathrm{kl}_+^{-1}(p',\varepsilon) \ge p' > q^*$, which is already strict. In Case 2 we split into two subcases. If $q^* = p'$, then $\mathrm{kl}(p'\|q^*) = \mathrm{kl}(p'\|p') = 0 < \varepsilon$; by continuity of $q \mapsto \mathrm{kl}(p'\|q)$ on $[p',1)$, there exists $q > q^*$ with $\mathrm{kl}(p'\|q) < \varepsilon$, so $\mathrm{kl}_+^{-1}(p',\varepsilon) \ge q > q^*$. If instead $q^* > p'$, then for fixed q^* the derivative $\frac{\partial}{\partial p}\mathrm{kl}(p\|q^*) = \ln\frac{p(1-q^*)}{q^*(1-p)}$ is strictly negative on $(0,q^*)$, so $r \mapsto \mathrm{kl}(r\|q^*)$ is strictly decreasing on the open interval $(0,q^*)$. The strict decrease extends to the left endpoint $r = 0$ as follows: for any $p_1 \in (0,p')$, the open-interval strict decrease gives $\mathrm{kl}(p'\|q^*) < \mathrm{kl}(p_1\|q^*)$, and the weak monotonicity from Case 2 gives $\mathrm{kl}(p_1\|q^*) \le \mathrm{kl}(0\|q^*)$, hence $\mathrm{kl}(p'\|q^*) < \mathrm{kl}(0\|q^*)$. Since $p < p' < q^*$, this yields $\mathrm{kl}(p'\|q^*) < \mathrm{kl}(p\|q^*) \le \varepsilon$ in both the $p > 0$ subcase (open-interval strict decrease applied directly) and the $p = 0$ subcase (sandwich just established). Again, by continuity, there exists $q > q^*$ with $\mathrm{kl}(p'\|q) < \varepsilon$, so $\mathrm{kl}_+^{-1}(p',\varepsilon) \ge q > q^*$. □

5 PAC-Bayes-kl Inequality

Theorem 5.1 (PAC-Bayes-kl for measurable posterior kernels [6, 4])**.** *Let (Θ, Σ_Θ) be a measurable hypothesis space. Let P be a prior on Θ independent of $S_1 \sim \mathcal{D}^{n_1}$, with $n_1 \ge 8$. Let Q be a stochastic kernel from samples S_1 to probability measures on Θ. Assume $((\theta,(x,y)) \mapsto \ell(h_\theta(x),y))$ is measurable and takes values in $[0,1]$. Then for any $\delta \in (0,1)$:*

$$\Pr_{S_1}\left[\mathrm{kl}\Big(\hat{L}_{S_1}(Q_{S_1}) \,\|\, L_{\mathcal{D}}(Q_{S_1})\Big) \le \frac{\mathrm{KL}(Q_{S_1}\|P) + \ln(2\sqrt{n_1}/\delta)}{n_1}\right] \ge 1-\delta.$$

Inverting the kl *constraint via kl_+^{-1} yields an explicit upper bound on $L_{\mathcal{D}}(Q)$.*

This is the standard stochastic-kernel PAC-Bayes inequality specialized to bounded i.i.d. losses. Rivasplata et al. [6] provide the measure-theoretic posterior-kernel framework, and Maurer [4] provides the exponential-moment bound that yields the constant $C(n_1,\delta) = \ln(2\sqrt{n_1}/\delta)$. We use this cited combination as the input PAC-Bayes inequality for the remainder of the proof.

6 Certificate Correctness

Theorem 6.1 (Analytic certificate envelope for posterior-selection rules)**.** *Assume $n_1 \ge 8$. For $\mathcal{D}^{|S_0|}$-almost every realization of S_0, assume: (i) $P = P(S_0)$ is a fixed prior on Θ independent of S_1; (ii) $S_1 \mapsto Q =$*

$\mathcal{Q}(S_0, S_1)$ is a measurable posterior-selection kernel; (iii) $\mathrm{KL}_{ub}(Q) \geq \mathrm{KL}(Q\|P)$ is a deterministic certified upper bound, measurable as a function of the realized computation. Assume either

$$\varepsilon_{mc} > 0, \qquad \delta_{mc} \in (0,1), \qquad m \geq \left\lceil \frac{\ln(2/\delta_{mc})}{2\varepsilon_{mc}^2} \right\rceil,$$

or else $\hat{L}_{S_1}(Q)$ is computed exactly, in which case $\varepsilon_{mc} = 0$, $\delta_{mc} = 0$, and $\hat{L}_{S_1}^{(m)}(Q) := \hat{L}_{S_1}(Q)$. For $\delta_{pb} \in (0,1)$, define

$$p_{\mathrm{cert}} := \min\Big\{1,\ \hat{L}_{S_1}^{(m)}(Q) + \varepsilon_{mc}\Big\}, \qquad \varepsilon_{pb}^{\mathrm{cert}}(Q) := \frac{\mathrm{KL}_{ub}(Q) + C(n_1, \delta_{pb})}{n_1},$$

$$B_{\mathrm{ana}}(Q) := \mathrm{kl}_+^{-1}\big(p_{\mathrm{cert}},\ \varepsilon_{pb}^{\mathrm{cert}}(Q)\big).$$

Then, for $\mathcal{D}^{|S_0|}$-almost every S_0, with probability at least $1 - \delta_{pb} - \delta_{mc}$ over S_1 and all certificate-side Monte Carlo randomness,

$$L_{\mathcal{D}}(Q) \leq B_{\mathrm{ana}}(Q).$$

Corollary 6.2 (Returned numerical certificate)**.** *Suppose a deterministic certified inversion routine (Assumption 7.1) returns $B_{\mathrm{cert}}(Q)$ satisfying*

$$B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{ana}}(Q) + \eta_{num}.$$

Define the formal slack envelope

$$B_{\mathrm{env}}(Q) := B_{\mathrm{ana}}(Q) + \eta_{num}.$$

On the same event as Theorem 6.1,

$$L_{\mathcal{D}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{env}}(Q).$$

If the inversion routine is randomized, an additional numerical failure budget δ_{num} may be included: the displayed sandwich condition is then required to hold on a numerical event of conditional probability at least $1 - \delta_{num}$, given (S_0, S_1) and all previous certificate-side randomness.

Proof of Theorem 6.1. Fix S_0 outside the null set on which the stated measurability or prior assumptions may fail. Conditional on this S_0, the prior $P = P(S_0)$ is fixed and independent of S_1, and $S_1 \mapsto Q = \mathcal{Q}(S_0, S_1)$ is a measurable posterior-selection kernel; therefore Theorem 5.1 applies conditionally on this S_0. Let

$$E_{pb} := \left\{ \mathrm{kl}\Big(\hat{L}_{S_1}(Q) \,\|\, L_{\mathcal{D}}(Q)\Big) \leq \frac{\mathrm{KL}(Q\|P) + C(n_1, \delta_{pb})}{n_1} \right\}.$$

By Theorem 5.1, $\Pr(E_{pb} \mid S_0) \geq 1 - \delta_{pb}$ for the selected posterior-selection rule $Q = \mathcal{Q}(S_0, S_1)$.

On E_{pb}, writing

$$\varepsilon_{pb}(Q) := \frac{\mathrm{KL}(Q\|P) + C(n_1, \delta_{pb})}{n_1}$$

we have

$$\mathrm{kl}\Big(\hat{L}_{S_1}(Q) \,\|\, L_{\mathcal{D}}(Q)\Big) \leq \varepsilon_{pb}(Q).$$

If $L_{\mathcal{D}}(Q) \geq \hat{L}_{S_1}(Q)$, then $L_{\mathcal{D}}(Q)$ is feasible in the definition of kl_+^{-1}, so

$$L_{\mathcal{D}}(Q) \leq \mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}(Q)\Big).$$

If $L_{\mathcal{D}}(Q) < \hat{L}_{S_1}(Q)$, then

$$L_{\mathcal{D}}(Q) \leq \hat{L}_{S_1}(Q) \leq \mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}(Q)\Big),$$

since $\mathrm{kl}_+^{-1}(p, \varepsilon) \geq p$ for all $p \in [0,1]$ and $\varepsilon \geq 0$. Thus in all cases,

$$L_{\mathcal{D}}(Q) \leq \mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}(Q)\Big).$$

Let E_{mc} be the event

$$\left|\hat{L}_{S_1}^{(m)}(Q) - \hat{L}_{S_1}(Q)\right| \le \varepsilon_{mc}.$$

In the exact-computation case, $\hat{L}_{S_1}^{(m)}(Q) = \hat{L}_{S_1}(Q)$, so E_{mc} holds surely and $\Pr(E_{mc}^c \mid S_0) = 0 = \delta_{mc}$. Otherwise, conditional on (S_0, S_1), the random variables

$$Z_j := \hat{L}_{S_1}(h_{\theta_j}), \qquad \theta_j \overset{\text{iid}}{\sim} Q = \mathcal{Q}(S_0, S_1),$$

are i.i.d., take values in $[0,1]$, and satisfy $\mathbb{E}[Z_j \mid S_0, S_1] = \hat{L}_{S_1}(Q)$. Hoeffding's inequality gives $\Pr(E_{mc}^c \mid S_0, S_1) \le \delta_{mc}$, and taking conditional expectation over S_1 yields $\Pr(E_{mc}^c \mid S_0) \le \delta_{mc}$.

By the deterministic upper-bound assumption,

$$\varepsilon_{pb}(Q) \le \varepsilon_{pb}^{\text{cert}}(Q),$$

so by monotonicity in ε,

$$\mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}(Q)\Big) \le \mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}^{\text{cert}}(Q)\Big).$$

On E_{mc},

$$\hat{L}_{S_1}(Q) \le \hat{L}_{S_1}^{(m)}(Q) + \varepsilon_{mc} \quad \text{and} \quad \hat{L}_{S_1}(Q) \le 1,$$

so

$$\hat{L}_{S_1}(Q) \le \min\Big\{1,\ \hat{L}_{S_1}^{(m)}(Q) + \varepsilon_{mc}\Big\} = p_{\text{cert}},$$

so by monotonicity in p,

$$\mathrm{kl}_+^{-1}\Big(\hat{L}_{S_1}(Q), \varepsilon_{pb}^{\text{cert}}(Q)\Big) \le \mathrm{kl}_+^{-1}\big(p_{\text{cert}}, \varepsilon_{pb}^{\text{cert}}(Q)\big).$$

Combining the two events yields the analytic claim

$$L_{\mathcal{D}}(Q) \le B_{\text{ana}}(Q)$$

on $E_{pb} \cap E_{mc}$, and the total failure probability is at most $\delta_{pb} + \delta_{mc}$. □

Proof of Corollary 6.2. On the event from Theorem 6.1, we have $L_{\mathcal{D}}(Q) \le B_{\text{ana}}(Q)$. By assumption,

$$B_{\text{ana}}(Q) \le B_{\text{cert}}(Q) \le B_{\text{ana}}(Q) + \eta_{num} = B_{\text{env}}(Q),$$

which yields

$$L_{\mathcal{D}}(Q) \le B_{\text{cert}}(Q) \le B_{\text{env}}(Q).$$

□

Remark 6.3 (Analytic envelope vs. returned numerical certificate)**.** *The analytic certificate is* $B_{\text{ana}}(Q) = \mathrm{kl}_+^{-1}(p_{\text{cert}}, \varepsilon_{pb}^{\text{cert}}(Q))$*. The quantity used for computed non-vacuity and computed ranking is the returned certified endpoint* $B_{\text{cert}}(Q)$*, because it is computable and, on the high-probability event, satisfies* $L_{\mathcal{D}}(Q) \le B_{\text{cert}}(Q)$*. The formal slack envelope* $B_{\text{env}}(Q) = B_{\text{ana}}(Q) + \eta_{num}$ *is only a proof device, used explicitly in Theorem 10.1. The three objects satisfy*

$$B_{\text{ana}}(Q) \le B_{\text{cert}}(Q) \le B_{\text{env}}(Q).$$

For non-vacuity of the actual returned risk certificate, the natural sufficient condition is simply $B_{\text{cert}}(Q) < 1$*. If one also wants to certify* $B_{\text{env}}(Q) < 1$ *from the returned value alone, a sufficient condition is* $B_{\text{cert}}(Q) + \eta_{num} < 1$*.*

7 Polynomial-Time Computability

Assumption 7.1 (Certified Bernoulli-kl inversion)**.** *There is a certified inversion routine which, given $p \in [0,1]$, finite $\varepsilon \geq 0$, and tolerance $\eta > 0$, returns $\widehat{q} \in [0,1]$ satisfying*

$$\mathrm{kl}_+^{-1}(p,\varepsilon) \leq \widehat{q} \leq \mathrm{kl}_+^{-1}(p,\varepsilon) + \eta.$$

In the exact-real arithmetic model, this is obtained by monotone bisection using the fact that $q \mapsto \mathrm{kl}(p\|q)$ is strictly increasing on $[p,1)$ when $p < 1$ (Lemma 4.2). In the bit model, all comparisons of the form $\mathrm{kl}(p\|q) \leq \varepsilon$ are performed with certified interval arithmetic and outward rounding, or by an equivalent certified numerical method. The running time is polynomial in the input bit length and in $\log(1/\eta)$.

Theorem 7.2 (Polynomial-time certificate evaluation)**.** *Assume Assumption 7.1. Let $T_{\mathrm{samp}}(Q)$ denote the cost of drawing one sample $\theta \sim Q$ from the representation of Q, let T_{fwd} denote the cost of evaluating one weight sample h_θ on the full evaluation dataset S_1 (i.e., n_1 forward passes), and let T_{KL} denote the cost of computing $\mathrm{KL}_{ub}(Q)$. For fixed posterior Q, a returned certificate $B_{\mathrm{cert}}(Q)$ satisfying $B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{env}}(Q)$ and therefore (on the high-probability event of Theorem 6.1) $L_{\mathcal{D}}(Q) \leq B_{\mathrm{cert}}(Q)$, is computable in time*

$$O\big(T_{\mathrm{KL}} + m \cdot \big(T_{\mathrm{samp}}(Q) + T_{\mathrm{fwd}}\big) + T_{\mathrm{inv}}(\eta_{num})\big)\,,$$

where m is the Monte Carlo sample count and $T_{\mathrm{inv}}(\eta_{num})$ is the cost of the certified kl-inversion routine, equal to $O(\log_2(1/\eta_{num}))$ in the exact-real bisection model. For a finite hyperparameter grid Λ with union-bound confidence and per-candidate quantities $T_{\mathrm{KL},\lambda}$, $T_{\mathrm{samp},\lambda}$, $T_{\mathrm{fwd},\lambda}$, m_λ, and $\eta_{num,\lambda}$:

$$O\left(\sum_{\lambda\in\Lambda}\Big(T_{\mathrm{KL},\lambda} + m_\lambda \cdot \big(T_{\mathrm{samp},\lambda} + T_{\mathrm{fwd},\lambda}\big) + T_{\mathrm{inv}}(\eta_{num,\lambda})\Big)\right).$$

Both are polynomial in the stated problem parameters provided each $T_{\mathrm{KL},\lambda}$, $T_{\mathrm{samp},\lambda}$, $T_{\mathrm{fwd},\lambda}$, and T_{inv} is polynomial in (d, n_1). All runtime bounds are interpreted in the standard bit model in which arithmetic and logarithms are evaluated with certified interval arithmetic and outward rounding. The required working precision is included in the input bit length and in the certified inversion cost $T_{\mathrm{inv}}(\eta_{num,\lambda})$, as specified in Assumption 7.1.

Proof. The returned numerical certificate $B_{\mathrm{cert}}(Q)$ for a single posterior Q (satisfying $B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{env}}(Q)$) is computed by the following three-phase procedure:

Phase 1: KL computation. Compute $\mathrm{KL}_{ub}(Q) \geq \mathrm{KL}(Q\|P)$. For the isotropic Gaussian construction $P = \mathcal{N}(w_{\mathrm{early}}, \sigma^2 I)$, $Q = \mathcal{N}(w_{\mathrm{final}}, \sigma^2 I)$, this is $\|w_{\mathrm{final}} - w_{\mathrm{early}}\|^2/(2\sigma^2)$: a single $O(d)$ operation.

Phase 2: Monte Carlo estimation. Draw m i.i.d. samples $\theta_1, \ldots, \theta_m \sim Q$ and compute $\hat{L}_{S_1}^{(m)}(Q) = \frac{1}{m}\sum_{j=1}^m \hat{L}_{S_1}(h_{\theta_j})$. Each draw θ_j can be generated at cost $T_{\mathrm{samp}}(Q)$, and each $\hat{L}_{S_1}(h_{\theta_j})$ requires a full pass over S_1 at cost T_{fwd}. Total: $O\big(m \cdot \big(T_{\mathrm{samp}}(Q) + T_{\mathrm{fwd}}\big)\big)$. The Hoeffding requirement $m \geq \ln(2/\delta_{mc})/(2\varepsilon_{mc}^2)$ is polynomial in $(1/\varepsilon_{mc}, \log(1/\delta_{mc}))$.

Phase 3: certified kl-inverse computation. Compute a certified upper endpoint $B_{\mathrm{cert}}(Q)$ for $B_{\mathrm{ana}}(Q) = \mathrm{kl}_+^{-1}(p_{\mathrm{cert}}, \varepsilon_{pb}^{\mathrm{cert}}(Q))$ such that $B_{\mathrm{ana}}(Q) \leq B_{\mathrm{cert}}(Q) \leq B_{\mathrm{ana}}(Q) + \eta_{num} = B_{\mathrm{env}}(Q)$. If $p_{\mathrm{cert}} = 1$, then $B_{\mathrm{ana}}(Q) = \mathrm{kl}_+^{-1}(1, \varepsilon_{pb}^{\mathrm{cert}}) = 1$ by Lemma 4.2, so the routine returns $B_{\mathrm{cert}}(Q) = 1$. If $p_{\mathrm{cert}} < 1$, then $q \mapsto \mathrm{kl}(p_{\mathrm{cert}}\|q)$ is continuous and strictly increasing on $[p_{\mathrm{cert}}, 1)$, with limit $+\infty$ as $q \uparrow 1$. By Assumption 7.1, the certified inversion routine returns $B_{\mathrm{cert}}(Q)$ within additive tolerance η_{num}; in the exact-real bisection model, this requires $\lceil \log_2(1/\eta_{num}) \rceil$ steps (each an $O(1)$ evaluation of kl). In the bit model, the comparisons $\mathrm{kl}(p_{\mathrm{cert}}\|q) \leq \varepsilon_{pb}^{\mathrm{cert}}$ must be performed with certified or outward-rounded interval arithmetic. Total: $T_{\mathrm{inv}}(\eta_{num}) = O(\log_2(1/\eta_{num}))$ in the exact-real model.

Combining: $O\big(T_{\mathrm{KL}} + m \cdot \big(T_{\mathrm{samp}}(Q) + T_{\mathrm{fwd}}\big) + T_{\mathrm{inv}}(\eta_{num})\big)$, polynomial in all parameters.

Extension to finite grids. For a finite candidate set Λ, apply the above to each $\lambda \in \Lambda$ with its own PAC-Bayes / Monte Carlo / numerical confidence allocation and its own quantities $T_{\mathrm{KL},\lambda}$, $T_{\mathrm{samp},\lambda}$, $T_{\mathrm{fwd},\lambda}$, m_λ, and $\eta_{num,\lambda}$. Total runtime:

$$O\left(\sum_{\lambda\in\Lambda}\Big(T_{\mathrm{KL},\lambda} + m_\lambda \cdot \big(T_{\mathrm{samp},\lambda} + T_{\mathrm{fwd},\lambda}\big) + \log_2(1/\eta_{num,\lambda})\Big)\right).$$

Remark (extension to compact continuous Λ). The finite-grid result extends to compact $\Lambda \subset \mathbb{R}^d$ via standard covering-number arguments, under additional regularity. Assuming $\lambda \mapsto \mathrm{KL}(Q_\lambda \| P)$ is L_K-Lipschitz, $\lambda \mapsto \hat{L}_{S_1}(Q_\lambda)$ (the exact Gibbs loss, i.e., $\varepsilon_{mc} = 0$) is L_L-Lipschitz, and $p_{\mathrm{cert}}(\lambda) < 1$ for all $\lambda \in \Lambda$, the analytic envelope map $\lambda \mapsto B_{\mathrm{ana}}(\lambda)$ is Lipschitz on Λ. If $\eta_{num}(\lambda)$ is constant (or Lipschitz), the formal slack envelope $B_{\mathrm{env}}(\lambda) = B_{\mathrm{ana}}(\lambda) + \eta_{num}(\lambda)$ is also Lipschitz. The derivative bounds follow from $\varepsilon_{pb}^{\mathrm{cert}}(Q_\lambda) \geq C(n_1, \delta_{pb})/n_1 > 0$ (ensuring kl_+^{-1} is evaluated away from its singularity at $\varepsilon = 0$) and $p_{\mathrm{cert}}(\lambda) < 1$ (ensuring $q^* < 1$). Denoting $R := \mathrm{diam}(\Lambda)$ and L_{env} a Lipschitz constant of $\lambda \mapsto B_{\mathrm{env}}(\lambda)$, an $\varepsilon_{\mathrm{net}}$-net of size $M \leq (2RL_{\mathrm{env}}/\varepsilon_{\mathrm{approx}})^d$ then covers Λ to accuracy $\varepsilon_{\mathrm{approx}}$, yielding polynomial total runtime for fixed d. A full proof of this extension is omitted; the main results (Theorems 6.1–11.2) use only finite grids. □

8 Comparison Theorem

We first establish Pinsker's inequality, then the Maurer constant dominance lemma, and finally the comparison chain.

Lemma 8.1 (Pinsker's inequality [5])**.** *For all $p, q \in [0,1]$: $\mathrm{kl}(p\|q) \geq 2(p-q)^2$.*

Proof. For fixed $q \in (0,1)$, define $f(p) = \mathrm{kl}(p\|q)$ for $p \in (0,1)$. Then $f(q) = 0$, $f'(q) = [\ln(p/q) - \ln((1-p)/(1-q))]_{p=q} = 0$, and

$$f''(p) = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

By AM-GM, $p(1-p) \leq 1/4$, so $f''(p) \geq 4$ everywhere on $(0,1)$. Using Taylor's theorem with integral remainder, for $p \geq q$:

$$f(p) = f(q) + f'(q)(p-q) + \int_q^p (p-t)\, f''(t)\, dt \;\geq\; 4 \int_q^p (p-t)\, dt \;=\; 2(p-q)^2,$$

and symmetrically for $p \leq q$. Hence $\mathrm{kl}(p\|q) \geq 2(p-q)^2$ for all $p, q \in (0,1)$. For $p \in \{0,1\}$ with $q \in (0,1)$, continuity of $p \mapsto \mathrm{kl}(p\|q)$ on $[0,1]$ under the boundary conventions of Section 3 gives the inequality by limits. For $q \in \{0,1\}$ with $p \neq q$, $\mathrm{kl}(p\|q) = +\infty$ and the inequality is immediate; for $p = q \in \{0,1\}$, both sides vanish. □

Lemma 8.2 (Strict Pinsker on the upper-tail branch)**.** *For every $p \in [0,1]$ and every $q \in [p,1]$ with $q \neq p$,*

$$\mathrm{kl}(p\|q) > 2(q-p)^2.$$

Proof. If $p < 1$, define

$$h_p(q) := \mathrm{kl}(p\|q) - 2(q-p)^2, \qquad q \in [p,1).$$

Then $h_p(p) = 0$, and for every $q \in (p,1)$,

$$h_p'(q) = -\frac{p}{q} + \frac{1-p}{1-q} - 4(q-p) = \frac{(q-p)(2q-1)^2}{q(1-q)}.$$

Since $q(1-q) > 0$ on $(p,1)$ and $q - p > 0$, we have $h_p'(q) \geq 0$, with equality only at $q = \frac{1}{2}$. Hence for every $q > p$,

$$h_p(q) = \int_p^q h_p'(t)\, dt > 0,$$

because the integrand is nonnegative and positive except at the isolated point $t = \frac{1}{2}$. Thus $\mathrm{kl}(p\|q) > 2(q-p)^2$ for all $p < 1$ and $q \in (p,1)$. For $q = 1$ and $p < 1$, $\mathrm{kl}(p\|1) = +\infty > 2(1-p)^2$ directly. The case $p = 1$ is vacuous because $q \in [p,1]$ forces $q = 1$. □

Lemma 8.3 (Maurer constant dominance)**.** *For all $n \geq 1$ and $\delta \in (0,1)$:*

$$C_{\mathrm{Ma}}(n,\delta) := \ln \frac{2\sqrt{n}}{\delta} \leq \ln \frac{2n}{\delta} =: C_{\mathrm{Mc}}(n,\delta).$$

Equality holds iff $n = 1$.

Proof. $C_{\mathrm{Mc}} - C_{\mathrm{Ma}} = \ln \frac{2n/\delta}{2\sqrt{n}/\delta} = \ln \sqrt{n} = \frac{1}{2} \ln n \geq 0$, with equality iff $n = 1$. □

Bound variants. We compare four certificate variants arising from two choices of constant and two inversion methods. With C_{Ma} and C_{Mc} as above, define

$$\varepsilon_{\mathrm{Ma}} := \frac{\mathrm{KL}(Q\|P) + C_{\mathrm{Ma}}(n_1, \delta)}{n_1}, \qquad \varepsilon_{\mathrm{Mc}} := \frac{\mathrm{KL}(Q\|P) + C_{\mathrm{Mc}}(n_1, \delta)}{n_1}.$$

The four bounds are:

$$B_{\mathrm{kl}}^{\mathrm{Ma}} := \mathrm{kl}_+^{-1}(\hat{L}_{S_1}(Q),\ \varepsilon_{\mathrm{Ma}}), \qquad B_{\surd}^{\mathrm{Ma}} := \hat{L}_{S_1}(Q) + \sqrt{\varepsilon_{\mathrm{Ma}}/2},$$
$$B_{\mathrm{kl}}^{\mathrm{Mc}} := \mathrm{kl}_+^{-1}(\hat{L}_{S_1}(Q),\ \varepsilon_{\mathrm{Mc}}), \qquad B_{\surd}^{\mathrm{Mc}} := \hat{L}_{S_1}(Q) + \sqrt{\varepsilon_{\mathrm{Mc}}/2}.$$

Theorem 8.4 (Comparison theorem)**.** *For any $p \in [0,1]$ and $\varepsilon \geq 0$:*

$$\mathrm{kl}_+^{-1}(p, \varepsilon) \leq p + \sqrt{\varepsilon/2},$$

with equality iff $\varepsilon = 0$. Consequently, for $\varepsilon > 0$ the kl-inversion bound is strictly tighter than the square-root relaxation:

$$B_{\mathrm{kl}}^{\mathrm{Ma}} \leq B_{\surd}^{\mathrm{Ma}} \leq B_{\surd}^{\mathrm{Mc}},$$

with strict first inequality for all $n_1 \geq 1$ (by Lemma 8.2, since $C_{\mathrm{Ma}}(n_1, \delta) = \ln(2\sqrt{n_1}/\delta) > 0$ for all $\delta \in (0,1)$ and $n_1 \geq 1$, hence $\varepsilon_{\mathrm{Ma}} > 0$), and strict second inequality for $n_1 \geq 2$ (by Lemma 8.3).

Remark 8.5 (Empirical non-vacuity)**.** *At theorem level, formal non-vacuity is carried by the explicit five-configuration ImageNet family in Section 12, where all five displayed returned certified endpoints satisfy $B_{\mathrm{cert}} < 1$. Since on the certificate event $L_{\mathcal{D}}(Q_\lambda) \leq B_{\mathrm{cert}}(\lambda)$, this is the directly computable non-vacuity condition. The larger MNIST and ImageNet sweeps remain supplementary plug-in evidence: all 80 ImageNet configurations and all 10 MNIST configurations yield plug-in $B_{\mathrm{kl}}^{\mathrm{Ma}} < 1$, with the tightest plug-in value $B = 0.147$ (ConvNeXt-Large).*

Remark 8.6 (Clipped square-root certificates)**.** *The strict comparison above is stated for the unclipped square-root expression $p + \sqrt{\varepsilon/2}$. If one instead reports the clipped value $\min\{1, p + \sqrt{\varepsilon/2}\}$, strictness can fail at $p = 1$ (both sides equal 1); the non-strict inequality remains valid after clipping.*

Proof. Step 1 (kl-inversion Pinsker bound). We show $\mathrm{kl}_+^{-1}(p, \varepsilon) \leq p + \sqrt{\varepsilon/2}$ for all $p \in [0,1]$ and $\varepsilon \geq 0$.

Let $q^* = \mathrm{kl}_+^{-1}(p, \varepsilon)$, so $\mathrm{kl}(p\|q^*) \leq \varepsilon$. By Pinsker's inequality (Lemma 8.1), $\mathrm{kl}(p\|q^*) \geq 2(p - q^*)^2$. Since $q^* \geq p$:

$$\varepsilon \geq \mathrm{kl}(p\|q^*) \geq 2(q^* - p)^2, \quad \text{so} \quad q^* - p \leq \sqrt{\varepsilon/2}, \quad \text{giving} \quad q^* \leq p + \sqrt{\varepsilon/2}.$$

Equality analysis. If $\varepsilon = 0$, then Lemma 4.2 gives $q^* = p$, so equality holds. For $\varepsilon > 0$ and $p \in (0,1)$: Lemma 4.2 gives $q^* \in (p, 1)$ with $\mathrm{kl}(p\|q^*) = \varepsilon$; by strict upper-tail Pinsker (Lemma 8.2), $\varepsilon = \mathrm{kl}(p\|q^*) > 2(q^* - p)^2$, hence $q^* < p + \sqrt{\varepsilon/2}$. The case $p = 0$ is also strict. Define

$$g(q) := -\ln(1 - q) - 2q^2, \qquad q \in [0, 1).$$

Then $g(0) = 0$ and

$$g'(q) = \frac{1}{1-q} - 4q = \frac{(2q-1)^2}{1-q} \geq 0,$$

with strict positivity except at $q = \frac{1}{2}$. Hence $g(q) > 0$ for every $q \in (0,1)$, i.e.

$$\mathrm{kl}(0\|q) = -\ln(1 - q) > 2q^2.$$

Therefore the Pinsker step is strict when $p = 0$ and $q^* > 0$. The case $p = 1$: the feasible set in the definition of $\mathrm{kl}_+^{-1}(1, \varepsilon)$ is $[1, 1] = \{1\}$ (see Lemma 4.2), so $\mathrm{kl}_+^{-1}(1, \varepsilon) = 1$; hence $\mathrm{kl}_+^{-1}(1, \varepsilon) = 1 < 1 + \sqrt{\varepsilon/2}$ for $\varepsilon > 0$. Note that this conclusion does not rely on any claim that $\mathrm{kl}(1\|q) = +\infty$ for $q < 1$; in fact $\mathrm{kl}(1\|q) = \ln(1/q)$ is finite on $(0, 1]$.

Step 2 (Comparison chain). Define:

$$\varepsilon_{\mathrm{Ma}} = \frac{\mathrm{KL}(Q\|P) + C_{\mathrm{Ma}}(n_1, \delta)}{n_1}, \qquad \varepsilon_{\mathrm{Mc}} = \frac{\mathrm{KL}(Q\|P) + C_{\mathrm{Mc}}(n_1, \delta)}{n_1}.$$

By Lemma 8.3, $C_{\mathrm{Ma}} \le C_{\mathrm{Mc}}$, so $\varepsilon_{\mathrm{Ma}} \le \varepsilon_{\mathrm{Mc}}$. By Theorem 4.3:

$$B_{\mathrm{kl}}^{\mathrm{Ma}} = \mathrm{kl}_+^{-1}(\hat{L}, \varepsilon_{\mathrm{Ma}}) \le \mathrm{kl}_+^{-1}(\hat{L}, \varepsilon_{\mathrm{Mc}}) = B_{\mathrm{kl}}^{\mathrm{Mc}}.$$

By Step 1 (Pinsker bound applied to $B_{\mathrm{kl}}^{\mathrm{Mc}}$):

$$B_{\mathrm{kl}}^{\mathrm{Mc}} = \mathrm{kl}_+^{-1}(\hat{L}, \varepsilon_{\mathrm{Mc}}) \le \hat{L} + \sqrt{\varepsilon_{\mathrm{Mc}}/2} = B_{\surd}^{\mathrm{Mc}}.$$

Applying Step 1 with Maurer's constant: $B_{\mathrm{kl}}^{\mathrm{Ma}} = \mathrm{kl}_+^{-1}(\hat{L}, \varepsilon_{\mathrm{Ma}}) \le \hat{L} + \sqrt{\varepsilon_{\mathrm{Ma}}/2} = B_{\surd}^{\mathrm{Ma}}$. Since $\varepsilon_{\mathrm{Ma}} \le \varepsilon_{\mathrm{Mc}}$: $B_{\surd}^{\mathrm{Ma}} = \hat{L} + \sqrt{\varepsilon_{\mathrm{Ma}}/2} \le \hat{L} + \sqrt{\varepsilon_{\mathrm{Mc}}/2} = B_{\surd}^{\mathrm{Mc}}$, with strict inequality for $n_1 \ge 2$. Hence the full chain:

$$B_{\mathrm{kl}}^{\mathrm{Ma}} \le B_{\surd}^{\mathrm{Ma}} \le B_{\surd}^{\mathrm{Mc}}.$$

Step 3 (Non-vacuity). The comparison chain is a purely analytic inequality; non-vacuity requires a concrete witness (P, Q, S, δ) for which the certified value is < 1. Section 12 provides this theorem-level witness via the explicit five-configuration family $\Lambda_{\mathrm{Img}}^{(5)}$, where all five displayed returned certified endpoints satisfy $B_{\mathrm{cert}} < 1$. The broader 80-configuration ImageNet sweep is retained as supplementary plug-in evidence, with tightest plug-in value $B = 0.147$ for ConvNeXt-Large. □

9 Kendall τ Lower Bound

Theorem 9.1 (Audit-certified Kendall tau lower bound)**.** *Let $T \sim \mathcal{D}^{n_2}$ be an audit sample independent of (S_0, S_1). Let U denote any auxiliary non-audit randomness independent of T conditional on (S_0, S_1), and let V denote any auxiliary audit-computation randomness independent of T conditional on (S_0, S_1, U). Let $\Lambda = \{\lambda_1, \dots, \lambda_M\}$ be a fixed finite family with $M \ge 2$. For each $\lambda \in \Lambda$, let B_λ be a certificate value measurable with respect to (S_0, S_1, U), and let $I_\lambda = [\ell_\lambda, u_\lambda]$ be an interval measurable with respect to (S_0, S_1, U, T, V). Assume there is an event E_{audit}, measurable with respect to (S_0, S_1, U, T, V), such that*

$$\Pr(E_{\mathrm{audit}} \mid S_0, S_1, U) \ge 1 - \alpha$$

and, on E_{audit},

$$L_{\mathcal{D}}(Q_\lambda) \in I_\lambda \qquad \text{for all } \lambda \in \Lambda.$$

To define τ_a in the presence of ties, fix deterministic total-order extensions of the certificate values and of the true-risk values; these tie-breaking rules must not depend on the audit sample T. Counted pairs below are strictly ordered by the audit intervals, so true-risk ties can only occur among uncounted pairs, which are pessimistically treated as discordant in the lower bound. Throughout, τ_a denotes Kendall's pairwise concordance score after applying this fixed deterministic tie-breaking rule.

Let

$$N := \binom{M}{2},$$

and let C be the number of unordered pairs $\{\lambda, \lambda'\} \subset \Lambda$ such that either

$$B_\lambda < B_{\lambda'} \quad \text{and} \quad u_\lambda < \ell_{\lambda'},$$

or

$$B_{\lambda'} < B_\lambda \quad \text{and} \quad u_{\lambda'} < \ell_\lambda.$$

Then on E_{audit},

$$\tau_a(B, L_{\mathcal{D}}) \ge \frac{2C}{N} - 1.$$

In particular, if $C = N$, then

$$\tau_a(B, L_{\mathcal{D}}) = 1.$$

Proof. On E_{audit}, every counted pair is certified concordant. Indeed, if

$$B_\lambda < B_{\lambda'} \quad \text{and} \quad u_\lambda < \ell_{\lambda'},$$

then

$$L_{\mathcal{D}}(Q_\lambda) \le u_\lambda < \ell_{\lambda'} \le L_{\mathcal{D}}(Q_{\lambda'}),$$

so the certificate order agrees with the true-risk order. The symmetric case

$$B_{\lambda'} < B_\lambda \quad \text{and} \quad u_{\lambda'} < \ell_\lambda$$

is identical.

Hence at least C pairs are concordant. In the worst case, we pessimize every uncounted pair to discordant (contribution -1) after the fixed deterministic tie-breaking; counted pairs contribute $+1$. This gives $\#\text{concordant} \ge C$ and $\#\text{discordant} \le N - C$. Since the total number of unordered pairs is N, Kendall's τ_a satisfies

$$\tau_a(B, L_{\mathcal{D}}) = \frac{\#\text{concordant} - \#\text{discordant}}{N} \ge \frac{C - (N - C)}{N} = \frac{2C}{N} - 1.$$

If $C = N$, then every pair is certified concordant, so $\tau_a(B, L_{\mathcal{D}}) = 1$. □

Definition 9.2 (Criterion 3 at threshold τ_0)**.** *Fix a finite family Λ with $|\Lambda| \ge 2$, and let $B_{\text{cert}} := (B_{\text{cert}}(Q_\lambda))_{\lambda\in\Lambda}$ denote the vector of returned certified numerical endpoints. We say that the predictive-tightness criterion is satisfied at threshold $\tau_0 \in [-1, 1]$ if*

$$\tau_a(B_{\text{cert}}, L_{\mathcal{D}}) \ge \tau_0.$$

For the explicit witness families in Section 12, the two thresholds used in the manuscript are

$$\tau_0^{(5)} := 0.8 \qquad \text{for } \Lambda_{\text{Img}}^{(5)}, \qquad \tau_0^{(4)} := 1 \qquad \text{for } \Lambda_{\text{Img}}^{(4)}.$$

10 Direct Monotonicity

For a hyperparameter grid Λ with associated posteriors $\{Q_\lambda\}_{\lambda\in\Lambda}$, write $\mathrm{KL}_{ub}(\lambda) := \mathrm{KL}_{ub}(Q_\lambda)$. When the numerical tolerance depends on λ, the formal slack envelope (Section 3) reads

$$B_{\text{env}}(\lambda) := \mathrm{kl}_+^{-1}\big(p_{\text{cert}}(\lambda),\, \varepsilon_{pb}^{\text{cert}}(Q_\lambda)\big) + \eta_{num}(\lambda).$$

This theorem concerns B_{env}. It does not by itself prove an ordering of the returned numerical endpoints B_{cert}, unless the numerical routine is also known to preserve the relevant ordering or the ordering is certified with an explicit separation margin; see the remark after the theorem.

Theorem 10.1 (Direct monotonicity of the formal envelope)**.** *Assume*

$$p_{\text{cert}}(\lambda) < p_{\text{cert}}(\lambda') \le 1, \qquad \varepsilon_{pb}^{\text{cert}}(Q_\lambda) \le \varepsilon_{pb}^{\text{cert}}(Q_{\lambda'}), \qquad \eta_{num}(\lambda) \le \eta_{num}(\lambda').$$

Then $B_{\text{env}}(\lambda) < B_{\text{env}}(\lambda')$.

Proof. Step 1 (Strict monotonicity in the first argument). Since $\mathrm{KL}_{ub}(\lambda) \ge 0$ and $C(n_1, \delta_{pb}) = \ln(2\sqrt{n_1}/\delta_{pb}) > 0$ for all $n_1 \ge 1$ and $\delta_{pb} \in (0, 1)$, we have $\varepsilon_{pb}^{\text{cert}}(Q_\lambda) = (\mathrm{KL}_{ub}(\lambda) + C(n_1, \delta_{pb}))/n_1 > 0$. Theorem 4.4 therefore gives

$$\mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda),\, \varepsilon_{pb}^{\text{cert}}(Q_\lambda)) < \mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda'),\, \varepsilon_{pb}^{\text{cert}}(Q_\lambda)).$$

Step 2 (Weak monotonicity in the second argument). By Theorem 4.3,

$$\mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda'),\, \varepsilon_{pb}^{\text{cert}}(Q_\lambda)) \le \mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda'),\, \varepsilon_{pb}^{\text{cert}}(Q_{\lambda'})).$$

Combining Steps 1 and 2 yields

$$\mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda), \varepsilon_{pb}^{\text{cert}}(Q_\lambda)) < \mathrm{kl}_+^{-1}(p_{\text{cert}}(\lambda'), \varepsilon_{pb}^{\text{cert}}(Q_{\lambda'})).$$

Adding $\eta_{num}(\lambda)$ to the left and $\eta_{num}(\lambda')$ to the right preserves the strict inequality because $\eta_{num}(\lambda) \le \eta_{num}(\lambda')$. Therefore $B_{\text{env}}(\lambda) < B_{\text{env}}(\lambda')$. □

Remark 10.2 (Returned endpoints and ordering margins)**.** *The returned endpoints satisfy* $B_{\text{ana}}(\lambda) \leq B_{\text{cert}}(\lambda) \leq B_{\text{env}}(\lambda)$*. An ordering of* B_{env} *does not automatically imply the same ordering of returned endpoints* B_{cert}*. In the master theorem and in Criterion 3, the Kendall ranking is taken directly with respect to the returned values* B_{cert}*. If one wants to infer an ordering of* B_{env} *from returned values alone, a sufficient condition is* $B_{\text{cert}}(\lambda) + \eta_{num}(\lambda) < B_{\text{cert}}(\lambda')$*. In the theorem-level witness family of Section 12,* $\eta_{num} = 10^{-10}$ *and the minimum separation between displayed* B_{cert} *values exceeds* 0.01*, so the orderings of* B_{ana}*,* B_{cert}*, and* B_{env} *all coincide.*

11 Master Theorem

Assumption 11.1 (Prior independence under the population idealization)**.** *Throughout, we adopt the standard ML-theory idealization that the available labeled data are i.i.d. draws from a population distribution* $\mathcal{D}$*, with realized splits* S_0*,* S_1*, and* T *pairwise disjoint (by a deterministic index split or by a random split independent of the data). The prior* $P_\lambda = P_\lambda(S_0)$ *is constructed from* S_0 *alone, or from public information whose generative process is independent of* (S_1, T) *— e.g., publicly distributed pretrained checkpoints whose training data are disjoint from* (S_1, T)*. In either case* P_λ *is measurable with respect to a* σ*-field independent of* (S_1, T)*, as required by the PAC-Bayes prior-independence convention. The empirical instantiation in Section 12 (Gaussian priors centered at publicly distributed ImageNet-1K checkpoints whose training set is disjoint from the seed-fixed* (S_0, S_1, T) *partition) is consistent with this assumption under the population/checkpoint-generation idealization above; the assumption itself is what the theorem requires, and we do not separately establish that every public-checkpoint engineering choice (architecture, preprocessing, hyperparameter grid) is uninfluenced by ImageNet-1K validation performance — that condition is part of the idealization.*

Theorem 11.2 (Certified sample-split Gibbs framework for all three criteria)**.** *Let* $T \sim \mathcal{D}^{n_2}$ *be an audit sample independent of* (S_0, S_1)*, and let* $\Lambda = \{\lambda_1, \ldots, \lambda_M\}$ *be a fixed finite family with* $M \geq 2$*. For each* $\lambda \in \Lambda$*, let:*

- $P_\lambda = P_\lambda(S_0)$ *be a prior independent of* (S_1, T)*,*
- $Q_\lambda = \mathcal{Q}_\lambda(S_0, S_1)$ *be a measurable posterior-selection rule,*
- $\mathrm{KL}_{ub}(\lambda) \geq \mathrm{KL}(Q_\lambda \| P_\lambda)$ *be a deterministic certified upper bound (matching the convention of Theorem 6.1),*
- $B_{\text{cert}}(\lambda)$ *be the returned certified numerical endpoint from Corollary 6.2, satisfying* $B_{\text{ana}}(\lambda) \leq B_{\text{cert}}(\lambda) \leq B_{\text{env}}(\lambda)$*,*
- $I_\lambda = [\ell_\lambda, u_\lambda]$ *be an audit interval satisfying the hypothesis of Theorem 9.1.*

Let U *denote the aggregate certificate-computation randomness appearing in the returned certificate values* $\{B_{\text{cert}}(\lambda) : \lambda \in \Lambda\}$*, and assume* $U \perp T \mid (S_0, S_1)$ *(by the setup convention of Section 3). Let* V *denote any auxiliary audit-computation randomness, and assume* $V \perp T \mid (S_0, S_1, U)$*. Assume there is an audit event* E_{audit}*, measurable with respect to* (S_0, S_1, U, T, V)*, such that*

$$\Pr(E_{\text{audit}} \mid S_0, S_1, U) \geq 1 - \alpha \qquad \text{and, on } E_{\text{audit}}, \quad L_{\mathcal{D}}(Q_\lambda) \in I_\lambda \text{ for all } \lambda \in \Lambda.$$

For each $\lambda \in \Lambda$*, allocate PAC-Bayes, Monte Carlo, and numerical failure budgets* $\delta_{pb,\lambda}$*,* $\delta_{mc,\lambda}$*, and* $\delta_{num,\lambda}$*. Let* α *be the audit failure budget, and define*

$$\Delta := \sum_{\lambda \in \Lambda} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big) + \alpha.$$

Assume all certificate computations and audit-interval computations are polynomial-time in the problem parameters. Then with probability at least $1 - \Delta$ *over* S_1*, the certificate-computation randomness* U*, the audit sample* T*, and the audit-interval computation randomness* V*, conditional on* S_0*, the following hold simultaneously:*

(I) ***Certified non-vacuity.*** *For every $\lambda \in \Lambda$,*

$$L_{\mathcal{D}}(Q_\lambda) \le B_{\mathrm{cert}}(\lambda).$$

Consequently, if the returned certified values satisfy

$$B_{\mathrm{cert}}(\lambda) < 1 \qquad \text{for all } \lambda \in \Lambda,$$

then the entire family is formally non-vacuous.

(II) ***Polynomial-time certificate evaluation.*** *All returned certificate values $B_{\mathrm{cert}}(\lambda)$ and all audit intervals I_λ are computable in polynomial time.*

(III) ***Certified predictive tightness.*** *Let C be the number of certified concordant pairs from Theorem 9.1, applied with $B_\lambda := B_{\mathrm{cert}}(\lambda)$. Then*

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) \ge \frac{2C}{\binom{|\Lambda|}{2}} - 1.$$

In particular, if every pair is certified concordant, then

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) = 1.$$

Therefore the realized family satisfies Criterion 3 at every threshold τ_0 with

$$\tau_0 \le \frac{2C}{\binom{|\Lambda|}{2}} - 1$$

in the sense of Definition 9.2.

Consequently, any explicit realized family Λ, fixed as in this theorem or selected under the hypotheses of Corollary 11.3 (random-family selection from a simultaneously-covered pool), for which all returned certified endpoints satisfy $B_{\mathrm{cert}}(\lambda) < 1$ and for which the audit-certified Kendall lower bound meets the chosen threshold τ_0, is a rigorous witness family for the three closure criteria.

Proof. For each $\lambda \in \Lambda$, let $E_{\mathrm{cert},\lambda}$ be the event on which Corollary 6.2 gives

$$L_{\mathcal{D}}(Q_\lambda) \le B_{\mathrm{cert}}(\lambda).$$

By Theorem 6.1 and Corollary 6.2,

$$\Pr(E^c_{\mathrm{cert},\lambda} \mid S_0) \le \delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}$$

for $\mathcal{D}^{|S_0|}$-almost every S_0. Let E_{audit} be the audit event from the theorem statement, and define

$$E := \Big(\bigcap_{\lambda\in\Lambda} E_{\mathrm{cert},\lambda}\Big) \cap E_{\mathrm{audit}}.$$

On E, Corollary 6.2 gives $L_{\mathcal{D}}(Q_\lambda) \le B_{\mathrm{cert}}(\lambda)$ for every $\lambda \in \Lambda$, proving part (I).

Part (II) is immediate from Theorem 7.2 together with the assumed polynomial-time computability of the audit intervals.

For part (III), Theorem 9.1 applies on E_{audit}, hence on E, and gives

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) \ge \frac{2C}{\binom{|\Lambda|}{2}} - 1.$$

Finally, by the union bound and the tower property,

$$\Pr(E^c \mid S_0) \le \sum_{\lambda\in\Lambda} \Pr(E^c_{\mathrm{cert},\lambda} \mid S_0) + \Pr(E^c_{\mathrm{audit}} \mid S_0).$$

By the theorem-statement hypothesis $\Pr(E_{\text{audit}} \mid S_0, S_1, U) \geq 1-\alpha$ and the tower property, $\Pr(E^c_{\text{audit}} \mid S_0) \leq \alpha$. Therefore

$$\Pr(E^c \mid S_0) \leq \sum_{\lambda \in \Lambda} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big) + \alpha = \Delta.$$

Therefore

$$\Pr(E \mid S_0) \geq 1 - \Delta.$$

□

Corollary 11.3 (Master Theorem with random-family selection)**.** *Let Λ_{pool} be a deterministic finite candidate family with simultaneous PAC-Bayes coverage as in Theorem 11.2 (i.e., the displayed PAC-Bayes confidence event holds for every $\lambda \in \Lambda_{\text{pool}}$ with total failure probability $\sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda}$). Let $\Lambda \subseteq \Lambda_{\text{pool}}$ be a (possibly random) finite subfamily selected by a rule measurable with respect to $(S_0, S_1, U_{\text{plug-in}})$, applied before the audit sample T is inspected; assume the audit-event hypothesis of Theorem 11.2 holds uniformly in the realization of the aggregate non-audit randomness $U = (U_{\text{plug-in}}, U_{\text{cert}})$, i.e., $\Pr(E_{\text{audit}} \mid S_0, S_1, U) \geq 1-\alpha$ a.s. Assume further that the certificate-side Monte Carlo randomness U_{cert} used to compute $\{B_{\text{cert}}(\lambda) : \lambda \in \Lambda\}$ is generated independently of the selection-rule randomness $U_{\text{plug-in}}$ (e.g., by drawing fresh i.i.d. samples after Λ is fixed); alternatively, certificate-side Monte Carlo coverage may itself be enforced simultaneously over Λ_{pool}, in which case U_{cert} may coincide with $U_{\text{plug-in}}$. Then the conclusions of Theorem 11.2 hold for Λ with probability at least*

$$1 - \sum_{\lambda \in \Lambda_{\text{pool}}} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big) - \alpha,$$

the deterministic worst-case lower bound obtained by union-bounding the certificate-side Monte Carlo and numerical events over the entire pool. Moreover, when U_{cert} is drawn fresh after selection (so the selection rule is independent of the cert-side MC), the certificate-side Monte Carlo and numerical sum is bounded by any deterministic constant D_{mc} satisfying $D_{\text{mc}} \geq \sum_{\lambda \in \Lambda}(\delta_{mc,\lambda} + \delta_{num,\lambda})$ almost surely; the corollary then holds with the (deterministic) tighter lower bound

$$1 - \sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda} - D_{\text{mc}} - \alpha,$$

via the conditional argument given in the proof. For uniform per-configuration budgets $\delta_{mc,\lambda} \equiv \delta_{mc}$, $\delta_{num,\lambda} \equiv \delta_{num}$ and any almost-sure cardinality bound $|\Lambda| \leq K$, one may take $D_{\text{mc}} = K(\delta_{mc} + \delta_{num})$.

Proof. For each $\lambda \in \Lambda_{\text{pool}}$, factor the per-$\lambda$ certificate event of Corollary 6.2 as $E_{\text{cert},\lambda} = E_{pb,\lambda} \cap E_{mc,\lambda} \cap E_{num,\lambda}$, where $E_{pb,\lambda}$ is the PAC-Bayes-kl coverage event (with $\Pr(E^c_{pb,\lambda} \mid S_0) \leq \delta_{pb,\lambda}$), $E_{mc,\lambda}$ is the certificate-side Monte Carlo concentration event (with $\Pr(E^c_{mc,\lambda} \mid S_0, S_1) \leq \delta_{mc,\lambda}$), and $E_{num,\lambda}$ is the numerical-bracket event (with $\Pr(E^c_{num,\lambda}) \leq \delta_{num,\lambda}$). On $E_{\text{cert},\lambda}$, $L_{\mathcal{D}}(Q_\lambda) \leq B_{\text{cert}}(\lambda)$. Define the *full-cert pool event*

$$E_{\text{cert,pool}} := \bigcap_{\lambda \in \Lambda_{\text{pool}}} E_{\text{cert},\lambda},$$

and the *PAC-Bayes-only pool event*

$$E_{pb,\text{pool}} := \bigcap_{\lambda \in \Lambda_{\text{pool}}} E_{pb,\lambda}.$$

By the simultaneous-pool PAC-Bayes hypothesis and a union bound over Λ_{pool},

$$\Pr\big(E^c_{pb,\text{pool}} \;\big|\; S_0\big) \;\leq\; \sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda},$$

and a further union bound over the MC and numerical events gives

$$\Pr\big(E^c_{\text{cert,pool}} \;\big|\; S_0\big) \;\leq\; \sum_{\lambda \in \Lambda_{\text{pool}}} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big).$$

Both events are measurable with respect to $\sigma(S_0, S_1, U)$, and on $E_{\text{cert,pool}}$ the certificate inequality $L_{\mathcal{D}}(Q_\lambda) \leq B_{\text{cert}}(\lambda)$ holds for every $\lambda \in \Lambda_{\text{pool}}$ simultaneously. Consequently it holds for every λ in any (S_0, S_1, U)-measurable subfamily $\Lambda \subseteq \Lambda_{\text{pool}}$, regardless of the random realization of Λ — the random selection rule cannot evict an already-covered configuration.

For the audit event, the hypothesis of Theorem 11.2 gives $\Pr(E_{\text{audit}} \mid S_0, S_1, U) \geq 1 - \alpha$ uniformly in the realization of (S_0, S_1, U). By the tower property,

$$\Pr\big(E^c_{\text{audit}} \;\big|\; S_0\big) \;=\; \mathbb{E}\big[\Pr(E^c_{\text{audit}} \mid S_0, S_1, U) \;\big|\; S_0\big] \;\leq\; \alpha.$$

A final union bound yields

$$\Pr\big(E^c_{\text{cert,pool}} \cup E^c_{\text{audit}} \;\big|\; S_0\big) \;\leq\; \sum_{\lambda \in \Lambda_{\text{pool}}} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big) + \alpha,$$

which is the deterministic worst-case lower bound stated in the corollary. On $E_{\text{cert,pool}} \cap E_{\text{audit}}$, conclusions (I)–(III) of Theorem 11.2 hold simultaneously for the realized $\Lambda \subseteq \Lambda_{\text{pool}}$.

For the tighter (random-Λ) lower bound, suppose the cert-side Monte Carlo randomness U_{cert} is drawn fresh after Λ is fixed — formally, conditional on the selection σ-field $\mathcal{F}_{\text{sel}} = \sigma(S_0, S_1, U_{\text{plug-in}})$ that determines Λ. On this conditioning, Λ is deterministic, and the Hoeffding and numerical events for Λ alone fail with conditional probability at most $\sum_{\lambda \in \Lambda}(\delta_{mc,\lambda} + \delta_{num,\lambda})$ over the fresh U_{cert}. PAC-Bayes coverage rides on the PAC-Bayes-only pool event $E_{pb,\text{pool}} = \bigcap_{\lambda \in \Lambda_{\text{pool}}} E_{pb,\lambda}$ defined above, with $\Pr(E^c_{pb,\text{pool}} \mid S_0) \leq \sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda}$, and this cost is $\mathcal{F}_{\text{sel}}$-measurable. Tower-integrating over $\mathcal{F}_{\text{sel}}$ and applying the deterministic dominator $D_{\text{mc}} \geq \sum_{\lambda \in \Lambda}(\delta_{mc,\lambda} + \delta_{num,\lambda})$ a.s. yields the deterministic tighter lower bound

$$1 - \sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda} - D_{\text{mc}} - \alpha,$$

matching the statement of the corollary. (Without invoking D_{mc}, the same tower step yields the equivalent expectation form $1 - \sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda} - \mathbb{E}[\sum_{\lambda \in \Lambda}(\delta_{mc,\lambda} + \delta_{num,\lambda})] - \alpha$.) For the realized witness, $|\Lambda| = 5$, $\delta_{mc,\lambda} \equiv 0.025$, and $\delta_{num,\lambda} \equiv 0$ (the certified bisection of Assumption 7.1 is deterministic, with bracket tolerance $\eta_{num,\lambda} \equiv 10^{-10}$ controlling the analytic-vs-returned envelope width but contributing zero to the failure-probability budget), so $D_{\text{mc}} = 5(0.025 + 0) = 0.125$ exactly is admissible. □

Remark 11.4 (No post-hoc audit-based family selection)**.** *If Λ is selected* after *inspecting audit intervals or audit losses computed from T, Corollary 11.3 does not apply. In that case, one must either include the entire candidate sweep in the simultaneous audit guarantee, or use a separate audit split for post-selection validation. The Section 12 family-selection protocol* avoids *post-hoc audit-based selection altogether: $\Lambda^{(5)}_{\text{Img}}$ is selected by an $(S_0, S_1, U_{\text{plug-in}})$-measurable rule from the deterministically-fixed* 80*-configuration pool Λ_{80}* before *the audit sample T is inspected, so the hypotheses of Corollary 11.3 are met directly.*

Remark 11.5 (Familywise confidence allocation)**.** Fixed-family case. *If Λ is deterministic and one targets familywise confidence $1 - \Delta_{\text{target}}$, it suffices to choose budgets satisfying*

$$\sum_{\lambda \in \Lambda} \big(\delta_{pb,\lambda} + \delta_{mc,\lambda} + \delta_{num,\lambda}\big) + \alpha \leq \Delta_{\text{target}}.$$

Random-family case. *If $\Lambda \subseteq \Lambda_{\text{pool}}$ is selected by an $(S_0, S_1, U_{\text{plug-in}})$-measurable rule from a deterministically pre-specified pool Λ_{pool} (as in Corollary 11.3 and the Section 12 witness), the relevant accounting is the pool-wide PAC-Bayes budget plus a deterministic post-selection MC/numerical dominator:*

$$\sum_{\lambda \in \Lambda_{\text{pool}}} \delta_{pb,\lambda} + D_{\text{mc}} + \alpha \leq \Delta_{\text{target}},$$

where $D_{\text{mc}} \geq \sum_{\lambda \in \Lambda}(\delta_{mc,\lambda} + \delta_{num,\lambda})$ almost surely is the deterministic upper bound from Corollary 11.3. Decreasing these budgets may enlarge the returned certified endpoints through the PAC-Bayes and Monte Carlo tolerances, so the certificate values should then be recomputed with the new allocation. The proof structure is unchanged.

12 Empirical Verification: explicit finite witness families and supplementary evidence

This section separates *theorem-level* claims from *supplementary plug-in* experiments. The theorem-level claim uses only configurations with fully tracked formal certificates in the sense of Theorem 6.1 together with an independent audit-set rank certification in the sense of Theorem 9.1. Larger experimental sweeps are reported separately as supplementary evidence and are not used as the core proof witness.

12.1 Explicit five-configuration ImageNet family

For the theorem-level claim we study the explicit five-configuration ImageNet family

$$\begin{aligned}\Lambda^{(5)}_{\mathrm{Img}} := \{&\text{ConvNeXt-Large, ViT-L/16, Swin-Tiny,}\\ &\text{EfficientNet-B0, ResNet-18}\} \times \{\sigma = 0.001,\ \text{no fine-tuning}\}.\end{aligned}$$

The open problem is existential, so it suffices to exhibit one explicit finite family satisfying the three criteria. The five configurations above are chosen as that displayed family: they span four architecture families and are the ImageNet-scale configurations for which this paper displays both fully tracked formal certificates and the audit-based predictive-tightness calculation.

Family-selection protocol. Here Λ_{80} denotes the deterministic pre-data 80-configuration sweep

$$\Lambda_{80} \;=\; \mathcal{A}_{10} \times \mathcal{S}_4 \times \mathcal{F}_2,$$

where the architecture, noise-scale, and fine-tune-flag axes are

$$\begin{aligned}\mathcal{A}_{10} \;&=\; \{\text{ResNet-18, ResNet-50, MobileNetV2, EfficientNet-B0, ViT-S/16,}\\ &\qquad \text{ViT-B/16, ViT-L/16, ConvNeXt-Tiny, ConvNeXt-Large, Swin-Tiny}\},\\ \mathcal{S}_4 \;&=\; \{\sigma \in \{0.001,\ 0.003,\ 0.005,\ 0.01\}\},\\ \mathcal{F}_2 \;&=\; \{\text{fine-tune, no-fine-tune}\},\end{aligned}$$

so $|\Lambda_{80}| = 10 \times 4 \times 2 = 80$ configurations, fixed before observing (S_0, S_1, T). The displayed family $\Lambda^{(5)}_{\mathrm{Img}}$ is selected by the following $(S_0, S_1, U_{\mathrm{plug-in}})$-measurable rule, applied before any audit computation on T: (i) take the best plug-in $B^{\mathrm{Ma}}_{\mathrm{kl}}$ representative per architecture family over Λ_{80} (yielding ConvNeXt-Large for Modern CNN, ViT-L/16 for Vision Transformer, Swin-Tiny for Swin, and EfficientNet-B0 for Classic CNN); ties are broken lexicographically by $(\sigma, \text{fine-tune flag}, \text{architecture key})$, with the lexicographic order itself fixed before observing the data (no actual ties occur in the realized values, by inspection); (ii) add ResNet-18 as a legacy Classic-CNN depth comparison. Step (i) is measurable with respect to $(S_0, S_1, U_{\mathrm{plug-in}})$ (the plug-in $B^{\mathrm{Ma}}_{\mathrm{kl}}$ values use $\hat{L}_{S_1}$, KL_{ub}, and the plug-in Monte Carlo draws $U_{\mathrm{plug-in}}$); step (ii) is a fixed architectural choice independent of all data. Thus the family indices $\{\lambda_1, \ldots, \lambda_5\}$ are measurable with respect to $\sigma(S_0, S_1, U_{\mathrm{plug-in}})$, while the formal certificates $B_{\mathrm{cert}}(\lambda)$ for $\lambda \in \Lambda^{(5)}_{\mathrm{Img}}$ are measurable with respect to $\sigma(S_0, S_1, U_{\mathrm{plug-in}}, U_{\mathrm{cert}})$, since they additionally depend on the cert-side Monte Carlo draws of paragraph (b) below. Both σ-fields are absorbed into the auxiliary non-audit randomness $U := (U_{\mathrm{plug-in}}, U_{\mathrm{cert}})$ of Theorem 9.1 and Corollary 11.3, where it is required only that U be independent of the audit sample T conditional on (S_0, S_1), which is satisfied by construction (cf. paragraph (c) below). The "fixed finite family" clause of Theorem 9.1 is therefore to be read conditional on $\sigma(S_0, S_1, U_{\mathrm{plug-in}})$ for the family indices and on $\sigma(S_0, S_1, U_{\mathrm{plug-in}}, U_{\mathrm{cert}})$ for the ranking values $B_{\mathrm{cert}}(\lambda)$; under either conditioning $\Lambda^{(5)}_{\mathrm{Img}}$ is deterministic, and under the latter the ranking values are deterministic as well.

Certificate-side pool coverage (S_1-side hardening). To discharge the analogous certificate-side concern that the same S_1 drives both family selection and the PAC-Bayes-kl inequality, we reserve simultaneous PAC-Bayes coverage for the entire 80-configuration sweep Λ_{80} via a Bonferroni union bound: allocating per-configuration PAC-Bayes budget $\delta_{pb,\lambda} = 0.125/80 \approx 1.5625 \times 10^{-3}$ uniformly over Λ_{80}, the

event "for every $\lambda \in \Lambda_{80}$ the PAC-Bayes-kl inequality of Theorem 5.1 holds" fails with probability at most $\sum_{\lambda \in \Lambda_{80}} \delta_{pb,\lambda} = 0.125$. This coincides with the PAC-Bayes sub-budget that an earlier allocation would have assigned to only the 5 displayed configurations, so the familywise confidence floor is unchanged. On this simultaneous-coverage event, every one of the 80 individual PAC-Bayes bounds is valid, so any $(S_0, S_1, U_{\text{plug-in}})$-measurable rule restricting the pool to a displayed subset inherits PAC-Bayes coverage without additional probability cost. The 5-member witness $\Lambda^{(5)}_{\text{Img}}$ is therefore a post-hoc $(S_0, S_1, U_{\text{plug-in}})$-measurable restriction of a uniformly-covered pool, and its S_1-dependent selection rule is harmless on the certificate side.

Formal specification of the Λ_{80} PAC-Bayes objects. The simultaneous PAC-Bayes coverage event of the previous paragraph is well-defined only if every $\lambda \in \Lambda_{80}$ carries an admissible PAC-Bayes triple $(P_\lambda, Q_\lambda, \text{KL}_{ub,\lambda})$. We make these objects explicit. For each architecture $a \in \mathcal{A}_{10}$, let w_a^{ref} denote the publicly distributed pretrained checkpoint; under Assumption 11.1, w_a^{ref} is measurable with respect to a σ-field independent of (S_0, S_1, T). For every $\lambda = (a, \sigma, f) \in \Lambda_{80}$ define the prior

$$P_\lambda \;=\; \mathcal{N}\big(w_a^{\text{ref}},\, \sigma^2 I\big)\,,$$

which is (S_0, S_1, T)-independent. The posterior selection rule is

$$Q_\lambda \;=\; \begin{cases} P_\lambda & \text{if } f = \text{no-fine-tune}, \\ \mathcal{N}\big(w_a^{\text{ref}} + v_\lambda(S_0),\, \sigma^2 I\big) & \text{if } f = \text{fine-tune}, \end{cases}$$

where $v_\lambda : S_0 \mapsto v_\lambda(S_0) \in \mathbb{R}^{\dim(w_a)}$ is a deterministic S_0-measurable mean shift produced by any fixed posterior-selection procedure operating on S_0 alone (e.g. a fixed number of SGD steps with deterministic initialization and optimizer schedule on S_0); the shift is independent of (S_1, T). The certified KL upper bound is then the closed form for two Gaussians sharing covariance $\sigma^2 I$,

$$\text{KL}_{ub,\lambda} \;=\; \begin{cases} 0 & \text{if } f = \text{no-fine-tune}, \\ \dfrac{\|v_\lambda(S_0)\|_2^2}{2\sigma^2} & \text{if } f = \text{fine-tune}, \end{cases}$$

which is the exact $\text{KL}(Q_\lambda \| P_\lambda)$ at the abstract-Gaussian level. For every $\lambda \in \Lambda_{80}$, the formal PAC-Bayes object is the continuous Gaussian distribution composed with the fixed deterministic finite-precision rounding map $\text{round} : \mathbb{R}^{\dim(w_a)} \to \mathcal{G}_{\text{fp}}$ used by the predictor h_θ during evaluation; the data-processing inequality applied to this rounding map yields $\text{KL}(\text{round}_* Q_\lambda \,\|\, \text{round}_* P_\lambda) \le \text{KL}(Q_\lambda \| P_\lambda) = \text{KL}_{ub,\lambda}$, so the continuous Gaussian KL is a certified upper bound for the implemented sampler at every pool element. For the selected no-fine-tune witnesses, $Q_\lambda = P_\lambda$ as continuous Gaussians, hence $\text{round}_* Q_\lambda = \text{round}_* P_\lambda$ and the implemented-distribution KL is exactly zero. The plug-in loss estimator used by step (i) of the family-selection rule is the standard sample-mean estimator $\hat{L}^{(m_{\text{plug}})}_{S_1}(Q_\lambda)$ over the plug-in MC draws $U_{\text{plug-in}}$; these plug-in values are not used in any formal certificate. The full Λ_{80} pool is invoked only for the simultaneous PAC-Bayes coverage event and family-selection measurability; the formal Monte Carlo-certified endpoints $B_{\text{cert}}(\lambda)$ of Theorem 6.1 are computed only for the five $\lambda \in \Lambda^{(5)}_{\text{Img}}$ selected by that protocol, all of which lie in the no-fine-tune slab $f = \text{no-fine-tune}$ at $\sigma = 0.001$ and therefore have $\text{KL}_{ub,\lambda} = 0$.

It is useful to distinguish three independent randomness sources used in the witness construction. *(a) Plug-in randomness* $U_{\text{plug-in}}$: the Monte Carlo draws used to compute the plug-in $B^{\text{Ma}}_{\text{kl}}$ values that drive step (i) of the selection rule. The plug-in MC values themselves do not require coverage: what is union-bounded over Λ_{80} is the PAC-Bayes-kl coverage event of Theorem 5.1, which is (S_0, S_1)-measurable and independent of $U_{\text{plug-in}}$. Hence any selection rule measurable in $(S_0, S_1, U_{\text{plug-in}})$ yields a subfamily of an already-covered pool. *(b) Certificate-side randomness* U_{cert}: the formal $m = 10$ Monte Carlo draws used in the displayed $\Lambda^{(5)}_{\text{Img}}$ certificates of Theorem 6.1, modeled as independent of $U_{\text{plug-in}}$ after selection (in the implementation, the realized stream is replayed by deterministic SHA-256 expansion of a logged master seed using a domain-separation tag disjoint from those of $U_{\text{plug-in}}$ and U_{audit}); the artifact log records the master seed and the per-shard derived integer seeds the implementation produces, cf. Appendix A. *(c) Audit-side randomness* U_{audit}: the per-witness m_a audit draws used by Theorem 9.1, modeled as independent of $(S_0, S_1, U_{\text{plug-in}}, U_{\text{cert}})$ (in

the implementation, the realized stream is replayed by deterministic SHA-256 seed expansion of the master seed using a domain-separation tag disjoint from those of $U_{\text{plug-in}}$ and U_{cert}); the artifact log records the master seed and the per-shard derived integer audit seeds (see Appendix A). Formally, U_{cert} and U_{audit} are modeled as ideal i.i.d. random variables generating the required draws from Q_λ, and the Hoeffding / empirical-Bernstein concentration inequalities are applied over the law of these random variables (equivalently, over the random base seed / random bit source from which the implementation derives them). The salted SHA-256 expansion of the realized master seed is a deterministic implementation detail used only to reproduce one realization of $(U_{\text{cert}}, U_{\text{audit}})$ for auditability; once the master seed is realized, the sample sequence is fixed and the concentration event has either occurred or failed, but the validity of the inequality is established before conditioning on the realized seed. Hence $U_{\text{audit}} \perp T \mid (S_0, S_1)$ as required by Theorem 9.1, and the audit-side conclusion is unaffected by the selection-side dependence.

Each configuration in $\Lambda^{(5)}_{\text{Img}}$ is evaluated with a fully tracked formal certificate using the sample-splitting setup of Section 3 and Theorem 6.1. For these five witnesses, the displayed Monte Carlo and numerical parameters are $m = 10$, $\delta_{mc} = 0.025$, and $\eta_{num} = 10^{-10}$; the PAC-Bayes budget is allocated uniformly across Λ_{80} via the Bonferroni union bound described above with $\delta_{pb,\lambda} = 1.5625 \times 10^{-3}$ per configuration.

For these five no-fine-tuning witnesses, the prior/posterior construction is explicit. For each architecture λ, let w^{ref}_λ denote the fixed pretrained reference weights used in the corresponding experiment, and define

$$P_\lambda = \mathcal{N}\big(w^{\text{ref}}_\lambda,\, \sigma^2 I\big) \qquad \text{with } \sigma = 0.001.$$

Because the witness regime uses no fine-tuning, the posterior-selection rule performs no S_1-dependent update and returns the same Gaussian:

$$Q_\lambda = P_\lambda.$$

For the present finite-precision interpretation of h_θ, $P_\lambda = Q_\lambda$ is read as the abstract continuous Gaussian composed with the fixed deterministic rounding map of the previous paragraph; equivalently, samples $\theta \sim P_\lambda$ are drawn at full precision and rounded to the implemented finite-precision grid before the forward pass. Since $Q_\lambda = P_\lambda$ as continuous Gaussians, the rounded distributions also coincide and $\mathrm{KL}(\mathrm{round}_* Q_\lambda \,\|\, \mathrm{round}_* P_\lambda) = 0$ exactly, so Theorem 6.1 applies on the implemented distribution with zero certified KL upper bound. Thus P_λ is independent of (S_1, T) — indeed it may be regarded as the measurable map $S_0 \mapsto P_\lambda$ that simply ignores S_0, provided the pretraining data used to produce the reference weights w^{ref}_λ is disjoint from (S_0, S_1, T). In our experiments, (S_0, S_1, T) is the seed-123 partition of the ImageNet-1K validation split (20,000/15,000/15,000 of the 50,000 images), while every w^{ref}_λ is pretrained on the disjoint ImageNet-1K training split ($\sim$ 1.28M images). Under the standard ML-theory idealization that ImageNet-1K is a sample from a population distribution, Assumption 11.1 renders w^{ref}_λ measurable with respect to a σ-field independent of (S_0, S_1, T), as required by the PAC-Bayes prior independence convention; we do not separately argue independence from disjointness of realized train/val index sets alone. Consequently

$$\mathrm{KL}(Q_\lambda \| P_\lambda) = 0 \qquad \text{for all } \lambda \in \Lambda^{(5)}_{\text{Img}}.$$

The displayed values are the returned certified endpoints $B_{\text{cert}}(\lambda)$ in the Section 3 convention $B_{\text{ana}}(\lambda) \le B_{\text{cert}}(\lambda) \le B_{\text{env}}(\lambda) = B_{\text{ana}}(\lambda) + \eta_{num}$; with $\eta_{num} = 10^{-10}$, the returned endpoint, the analytic envelope B_{ana}, and the formal slack envelope B_{env} agree to ten decimal places, and the displayed orderings are identical for the three.

The resulting returned certified endpoints are:

Architecture	Family	B_{cert}	Audit-risk estimate
ConvNeXt-Large	Modern CNN	0.622	0.133
ViT-L/16	Vision Transformer	0.632	0.142
Swin-Tiny	Swin	0.675	0.189
EfficientNet-B0	Classic CNN	0.711	0.223
ResNet-18	Classic CNN	0.778	0.286

Thus every member of $\Lambda^{(5)}_{\text{Img}}$ satisfies

$$B_{\text{cert}}(\lambda) < 1, \qquad \lambda \in \Lambda^{(5)}_{\text{Img}},$$

where $B_{\text{cert}}(\lambda)$ denotes the returned certified numerical endpoint (Section 3), so the family gives five explicit ImageNet-scale formal non-vacuity witnesses.

Per-witness numerical certificate. The intermediate values used by Theorem 6.1 and Corollary 6.2 to produce the B_{cert} values above are recorded below, enabling a self-contained external verification of every displayed endpoint. Across all five witnesses, $\mathrm{KL}_{ub}(Q_\lambda) = 0$ (since $Q_\lambda = P_\lambda$), $C(n_1, \delta_{pb,\lambda}) = \ln\big(2\sqrt{n_1}/\delta_{pb,\lambda}\big) \approx 11.9625$, $\varepsilon_{pb}^{\text{cert}} = C/n_1 \approx 7.975 \times 10^{-4}$, and $\varepsilon_{mc} = \sqrt{\ln(2/\delta_{mc})/(2m)} \approx 0.46808$.

λ	$\hat{L}_{S_1}^{(m)}(Q_\lambda)$	p_{cert}	B_{ana}	B_{cert}
ConvNeXt-Large	0.13483	0.60292	0.62234	0.62234
ViT-L/16	0.14505	0.61314	0.63246	0.63246
Swin-Tiny	0.18789	0.65597	0.67477	0.67477
EfficientNet-B0	0.22489	0.69298	0.71119	0.71119
ResNet-18	0.29356	0.76164	0.77838	0.77838

For each λ, $p_{\text{cert}} = \hat{L}_{S_1}^{(m)}(Q_\lambda) + \varepsilon_{mc}$ (Section 3), $B_{\text{ana}} = \mathrm{kl}_+^{-1}\big(p_{\text{cert}}, \varepsilon_{pb}^{\text{cert}}\big)$ is computed by the certified bisection of Assumption 7.1, and the returned numerical endpoint B_{cert} satisfies the certificate-safe one-sided envelope $B_{\text{ana}} \le B_{\text{cert}} \le B_{\text{ana}} + \eta_{num}$ with $\eta_{num} = 10^{-10}$, hence agrees with B_{ana} to five displayed decimals. Combining the table with the common $\varepsilon_{pb}^{\text{cert}}$ and ε_{mc} above, every step of $B_{\text{cert}} < 1$ is independently checkable from this document alone.

Remark 12.1 (Structural role of KL = 0 on this witness family)**.** *Because the witness family uses no fine-tuning ($Q_\lambda = P_\lambda$), $\mathrm{KL}(Q_\lambda \| P_\lambda) = 0$ for every $\lambda \in \Lambda_{\text{Img}}^{(5)}$, and the PAC-Bayes complexity slack reduces to the architecture-independent constant*

$$\varepsilon_{pb}^{\text{cert}} = C(n_1, \delta_{pb,\lambda})/n_1.$$

Ranking preservation on $\Lambda_{\text{Img}}^{(5)}$ is therefore a monotone consequence of a constant offset applied to the p_{cert}-induced ordering; the PAC-Bayes complexity term does no architecture-dependent work on this family. The theorem-level witness satisfies Criterion 3 in the existential sense asked by the open problem. A witness family in which the PAC-Bayes complexity term does architecture-dependent work (finite-KL fine-tuning, with a genuine KL/risk tradeoff across λ) is a separate and stronger program, noted in the Conclusion.

Audit-set intervals. For predictive-tightness certification we use an independent audit sample $T \sim \mathcal{D}^{n_2}$ of size $n_2 = 15{,}000$, disjoint from the certificate split (S_0, S_1). Set $M = 5$ and familywise audit level $\alpha = 0.05$. By Hoeffding's inequality with Bonferroni correction,

$$t = \sqrt{\frac{\ln(2M/\alpha)}{2n_2}} = \sqrt{\frac{\ln(200)}{30{,}000}} \approx 0.01329.$$

If the audit Gibbs losses $\hat{L}_T(Q_\lambda)$ are computed exactly (illustrative; not used for the present Gaussian witness family — see "Audit Monte Carlo certification" below), the simultaneous audit intervals are

$$I_\lambda = \Big[\max\{0, \hat{L}_T(Q_\lambda) - t\}, \min\{1, \hat{L}_T(Q_\lambda) + t\}\Big].$$

Monte Carlo audit Bonferroni split. When the audit Gibbs losses are Monte Carlo approximated, the total audit failure budget α must be split between the Hoeffding concentration of the n_2 audit samples (each $\hat{L}_T(Q_\lambda)$ around $L_{\mathcal{D}}(Q_\lambda)$) and the Monte Carlo concentration of the m_a posterior samples (each $\hat{L}_T^{(m_a)}(Q_\lambda)$ around $\hat{L}_T(Q_\lambda)$). Choose $\alpha_H > 0$ and $\{\delta_{\text{audit,mc},\lambda}\}_{\lambda\in\Lambda}$ with

$$\alpha_H + \sum_{\lambda\in\Lambda} \delta_{\text{audit,mc},\lambda} \le \alpha,$$

and replace the Hoeffding radius by $t_H = \sqrt{\ln(2M/\alpha_H)/(2n_2)}$. Let

$$\hat{L}_T^{(m_a)}(Q_\lambda) := \frac{1}{m_a}\sum_{j=1}^{m_a} \hat{L}_T(h_{\theta_{\lambda,j}}), \qquad \theta_{\lambda,j} \overset{\text{iid}}{\sim} Q_\lambda,$$

and pick m_a large enough that a valid concentration inequality for the bounded Monte Carlo draws (for example Hoeffding's inequality, or a variance-aware bound such as the two-sided empirical Bernstein inequality of [1]) gives, for each $\lambda \in \Lambda$, a certified half-width $\varepsilon_{\text{audit,mc}}(\lambda)$ satisfying

$$\Pr\Big(\Big|\hat{L}_T^{(m_a)}(Q_\lambda) - \hat{L}_T(Q_\lambda)\Big| \le \varepsilon_{\text{audit,mc}}(\lambda)\Big) \ge 1 - \delta_{\text{audit,mc},\lambda}.$$

By union bound over the M Hoeffding events and the M Monte Carlo events, the simultaneous audit event has failure probability at most $\alpha_H + \sum_\lambda \delta_{\text{audit,mc},\lambda} \le \alpha$. Then use the padded simultaneous intervals

$$I_\lambda = \Big[\max\Big\{0,\, \hat{L}_T^{(m_a)}(Q_\lambda) - t_H - \varepsilon_{\text{audit,mc}}(\lambda)\Big\}, \min\Big\{1,\, \hat{L}_T^{(m_a)}(Q_\lambda) + t_H + \varepsilon_{\text{audit,mc}}(\lambda)\Big\}\Big].$$

(In the exact-audit regime, $\alpha_H = \alpha$ and $\delta_{\text{audit,mc},\lambda} = 0$, so $t_H = t$ and the padded intervals reduce to the unpadded Hoeffding intervals above.)

Using the audit-risk estimates in the table above and no additional audit Monte Carlo padding, the intervals are:

$$I_{\text{CXL}} = [0.1207,\, 0.1473], \qquad I_{\text{ViT-L}} = [0.1287,\, 0.1553],$$
$$I_{\text{Swin}} = [0.1757,\, 0.2023], \qquad I_{\text{EffB0}} = [0.2097,\, 0.2363],$$
$$I_{\text{Res18}} = [0.2727,\, 0.2993].$$

The certificate ordering is

$$0.622 < 0.632 < 0.675 < 0.711 < 0.778,$$

which matches the ordering of the audit-risk estimates

$$0.133 < 0.142 < 0.189 < 0.223 < 0.286.$$

Among the 10 unordered pairs, every pair is certified concordant except

$$\{\text{ConvNeXt-Large},\ \text{ViT-L/16}\},$$

whose audit intervals overlap. Therefore $C = 9$ pairs are certified concordant. By Theorem 9.1,

$$\tau_a(B_{\text{cert}}, L_{\mathcal{D}}) \ge \frac{2C}{\binom{5}{2}} - 1 = \frac{18}{10} - 1 = 0.8.$$

Audit Monte Carlo certification. For the present Gaussian witness family the exact-audit regime is not practically available, so the theorem-level conclusion is discharged via the Monte Carlo regime with the two-sided empirical Bernstein concentration of [1]. The pre-declared per-witness draw counts are

$$m_a(\lambda) \in \{1{,}625,\ 2{,}380,\ 8{,}400,\ 7{,}820,\ 1{,}400\}$$

for λ in (ConvNeXt-Large, ViT-L/16, Swin-Tiny, EfficientNet-B0, ResNet-18) respectively, fixed before any audit observation; the per-witness budget is $\delta_{\text{audit,mc},\lambda} = 0.005$ and the Hoeffding budget is $\alpha_H = 0.025$, giving $\alpha_H + \sum_\lambda \delta_{\text{audit,mc},\lambda} = 0.05 = \alpha$ and

$$t_H \;=\; \sqrt{\frac{\ln(2M/\alpha_H)}{2n_2}} \;=\; 0.014132.$$

The empirical Bernstein bound takes the explicit form[1]

$$\varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\lambda) \;=\; \sqrt{\frac{2s_a^2(\lambda)\,\ln(3/\delta_{\mathrm{audit,mc},\lambda})}{m_a(\lambda)}} \;+\; \frac{3\ln(3/\delta_{\mathrm{audit,mc},\lambda})}{m_a(\lambda)},$$

with $s_a^2(\lambda)$ the Bessel-corrected sample variance of the audit MC draws. The realized values are:

λ	$\hat{L}_T^{(m_a)}(Q_\lambda)$	$m_a(\lambda)$	$\varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\lambda)$	I_λ
ConvNeXt-Large	0.13279	1,625	1.185×10^{-2}	[0.1068, 0.1588]
ViT-L/16	0.14174	2,380	8.11×10^{-3}	[0.1195, 0.1640]
Swin-Tiny	0.18928	8,400	2.31×10^{-3}	[0.1728, 0.2057]
EfficientNet-B0	0.22343	7,820	2.48×10^{-3}	[0.2068, 0.2400]
ResNet-18	0.28632	1,400	1.43×10^{-2}	[0.2579, 0.3147]

Direct pairwise check on the unrounded eB-padded intervals certifies $C = 9$ concordant pairs; the only overlapping pair is {ConvNeXt-Large, ViT-L/16}, exactly as anticipated from the certificate ordering. The bottleneck pair Swin-Tiny vs. EfficientNet-B0 separates by

$$\inf I_{\mathrm{EffB0}} \;-\; \sup I_{\mathrm{Swin}} \;=\; 0.20682 \;-\; 0.20572 \;=\; 1.10 \times 10^{-3} \;>\; 0,$$

and the remaining eight certified pairs separate by margin $\geq 8.85 \times 10^{-3}$. Therefore $C = 9$ on the unrounded eB-padded intervals, and Theorem 9.1 gives

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) \;\geq\; \frac{2C}{\binom{5}{2}} - 1 \;=\; \frac{18}{10} - 1 \;=\; 0.8$$

in the Monte Carlo regime for $\Lambda^{(5)}_{\mathrm{Img}}$.

Perfect-order deterministic-restriction corollary. On the same audit event E_{audit} that delivers the simultaneous coverage $1-\alpha$ for $\Lambda^{(5)}_{\mathrm{Img}}$, restricting attention to the 4-configuration subset

$$\begin{aligned}\Lambda^{(4)}_{\mathrm{Img}} := {}& \{\text{ConvNeXt-Large, Swin-Tiny, EfficientNet-B0, ResNet-18}\} \\ & \times\ \{\sigma = 0.001,\ \text{no fine-tuning}\}\end{aligned}$$

yields $\binom{4}{2} = 6$ unordered pairs, all of which are certified concordant on E_{audit}; hence

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}})\big|_{\Lambda^{(4)}_{\mathrm{Img}}} = 1$$

on the same audit event. This is a deterministic restriction of the $\Lambda^{(5)}_{\mathrm{Img}}$ audit conclusion (the choice of subset is deterministic once the audit intervals have been observed) and inherits the familywise confidence $1-\alpha$ without additional probability cost; it is not a separate pre-registered theorem-level witness family but a post-hoc corollary on the already-covered event. The unrounded eB-padded intervals certify all six pairwise separations within $\Lambda^{(4)}_{\mathrm{Img}}$ directly (smallest margin 1.10×10^{-3} at Swin-Tiny vs. EfficientNet-B0; remaining five margins $\geq 1.40 \times 10^{-2}$).

[1] We use the two-sided empirical Bernstein form of [1], Theorem 1 (eq. 7): for $X_1, \ldots, X_t$ i.i.d. in $[0, b]$ with sample mean $\bar{X}$ and MLE variance $V_t = (1/t)\sum_{i=1}^{t}(X_i - \bar{X})^2$, $\Pr\big(|\bar{X} - \mathbb{E}X| \leq \sqrt{2V_t x/t} + 3bx/t\big) \geq 1 - 3e^{-x}$; setting $x = \ln(3/\delta)$ and $b = 1$ recovers the displayed constants $(\ln(3/\delta), 3)$. The factor 3 in $1 - 3e^{-x}$ arises from a 3-event Bonferroni split (variance estimate, upper tail, lower tail), each at failure probability $e^{-x} = \delta/3$. Substituting the Bessel-corrected $s_a^2(\lambda) = (1/(m_a - 1))\sum_j (Z_j - \bar{Z})^2$ for V_t is conservative ($s_a^2 \geq V_t$), so the displayed half-width remains a valid upper bound.

Existential witness branch. Because the open problem is existential, $\Lambda^{(5)}_{\mathrm{Img}}$ serves as the theorem-level witness; the audit-side separation condition is discharged by the audit run reported above, yielding Criterion 3 at threshold $\tau_0^{(5)} = 0.8$, and the $\Lambda^{(4)}_{\mathrm{Img}}$ restriction then gives the sharper observational value $\tau_a = 1$ on the same audit event (Definition 9.2 threshold $\tau_0^{(4)} = 1$) as a corollary, not as a separate pre-registered witness. The original open-problem statement asks for predictive tightness in the rank-correlation sense ("strongly track"); we instantiate this open-ended requirement here by the finite-family Kendall threshold $\tau_0^{(5)} = 0.8$, which is one reasonable formalization rather than the unique reading of the problem statement.

Here "predictive tightness" is interpreted in the rank-correlation sense permitted by the open problem ("for example, high rank correlation"); the absolute magnitude gap $B_{\mathrm{cert}} - L_{\mathcal{D}}$ is dominated by ε_{mc} at $m = 10$ and tightens toward the plug-in value as $m \to \infty$, as documented in Remark 12.2.

Probability accounting. The PAC-Bayes sub-budget is paid once per configuration in Λ_{80} under the Bonferroni union bound, while the Monte Carlo and numerical sub-budgets are paid once per configuration in the displayed $\Lambda^{(5)}_{\mathrm{Img}}$, and the audit sub-budget α is familywise. The conjunction of simultaneous PAC-Bayes coverage over Λ_{80}, the Monte Carlo and numerical events on the displayed five, and the audit event holds with probability at least

$$1 - \sum_{\lambda \in \Lambda_{80}} \delta_{pb,\lambda} - \sum_{\lambda \in \Lambda^{(5)}_{\mathrm{Img}}} \left(\delta_{mc,\lambda} + \delta_{num,\lambda}\right) - \alpha.$$

With the allocations $\delta_{pb,\lambda} = 1.5625 \times 10^{-3}$ uniformly over Λ_{80} (total 0.125), $\delta_{mc,\lambda} = 0.025$ per displayed witness (total 0.125), $\delta_{num,\lambda} = 0$, and $\alpha = 0.05$, this gives a simultaneous confidence lower bound of $1 - 0.125 - 0.125 - 0 - 0.05 = 0.70$.

Pre-specified alternative allocation at 95% global confidence. Independently of the 70% allocation above, one may instead pre-specify the following audit-heavy allocation before running the certificate and audit computation, achieving global confidence target $\Delta_{\mathrm{target}} = 0.05$:

$$\sum_{\lambda \in \Lambda_{80}} \delta_{pb,\lambda} = 0.005, \qquad \sum_{\lambda \in \Lambda^{(5)}_{\mathrm{Img}}} \delta_{mc,\lambda} = 0.005, \qquad \alpha = 0.040,$$

with audit-side split $\alpha_H = 0.030$ and $\delta_{\mathrm{audit,mc},\lambda} = 0.002$ uniformly. Per configuration this gives $\delta_{pb,\lambda} = 6.25 \times 10^{-5}$ and $\delta_{mc,\lambda} = 10^{-3}$; the Maurer constant becomes $C(n_1, \delta_{pb,\lambda}) \approx 15.18$, the certificate-side Hoeffding tolerance $\varepsilon_{mc} \approx 0.6165$, and the audit Hoeffding half-width $t_H \approx 0.01392$. The recomputed returned certified endpoints are

$$B_{\mathrm{cert}}(\lambda) \in \{0.770,\ 0.780,\ 0.822,\ 0.857,\ 0.922\}$$

for $\lambda \in$ (ConvNeXt-Large, ViT-L/16, Swin-Tiny, EfficientNet-B0, ResNet-18), all strictly below 1. The recomputed empirical-Bernstein half-widths at $\delta_{\mathrm{audit,mc},\lambda} = 2 \times 10^{-3}$ give $\varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\text{Swin-Tiny}) \approx 2.64 \times 10^{-3}$ and $\varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\text{EfficientNet-B0}) \approx 2.84 \times 10^{-3}$, so the bottleneck audit budget is

$$2t_H + \varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\text{Swin-Tiny}) + \varepsilon^{\mathrm{eB}}_{\mathrm{audit,mc}}(\text{EfficientNet-B0}) \approx 0.03330,$$

which still falls strictly below the unrounded raw audit-mean gap 0.03415 between Swin-Tiny and EfficientNet-B0 by margin $+0.85 \times 10^{-3}$. Hence $C = 9$ remains certified, $\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) \geq 0.8$ holds, and the conclusions (I)–(III) of Theorem 11.2 obtain with $\Delta = 0.05$ in place of $\Delta = 0.30$. Both witnesses (the 70% pre-specified allocation above and the 95% pre-specified allocation here) discharge the open problem at confidence $1 - \delta$ for the displayed δ under their respective pre-data choices of the budget split. As a descriptive (post-hoc) observation about the realized audit data, the remaining headroom would permit any pre-specified global confidence up to approximately 96.8% at this m_a allocation, dominated by the relatively large sample variance of the ResNet-18 audit losses; pre-specifying confidence levels above this descriptive ceiling would require correspondingly larger per-witness m_a. We do not make a theorem-level claim about a confidence-level chosen post-hoc as the maximum the data happens to support.

Polynomial-time computation. Criterion 2 is provided by Theorem 7.2: for each fixed configuration, the formal certificate is computable in polynomial time in the stated problem parameters. Empirically, the observed certificate-evaluation times on the witness family were below 1,347s per ImageNet configuration on an NVIDIA RTX 3090. These wall-clock timings are reported as implementation evidence only; the theorem-level complexity claim is the polynomial-time certificate-evaluation result of Section 7.

12.2 Supplementary plug-in evidence

Beyond the theorem-level witness families above, we also ran larger plug-in sweeps on MNIST and ImageNet. In these supplementary experiments we set $\varepsilon_{mc} = 0$ and $\eta_{num} = 0$ and therefore report *plug-in approximations*, not formal certificates in the sense of Theorem 6.1.

MNIST. For the 784-600-600-10 network ($d = 837{,}610$), all 10/10 plug-in PAC-Bayes-kl values were non-vacuous, with best value 0.323. Within the reported group at $\sigma = 0.05$, the observed Kendall correlation between plug-in certificate values and audit-risk estimates was +1.0.

ImageNet. Across the larger ImageNet sweep (10 architectures, 80 configurations), all 80/80 plug-in PAC-Bayes-kl values were non-vacuous. The best plug-in values per architecture were:

Architecture	d	$B_{\mathrm{kl}}^{\mathrm{Ma}}$ (plug-in)	Gap
ConvNeXt-Large	197.8M	**0.147**	1.3%
ViT-L/16	304.3M	**0.157**	1.5%
ViT-B/16	86.6M	**0.165**	1.6%
ConvNeXt-Tiny	28.6M	**0.182**	0.6%
ViT-S/16	22.1M	**0.201**	1.5%
Swin-Tiny	28.3M	**0.201**	1.2%
EfficientNet-B0	5.3M	**0.239**	1.6%
MobileNetV2	3.5M	**0.287**	1.8%
ResNet-18	11.7M	**0.304**	1.8%
ResNet-50	25.6M	0.999	(degenerate)

These larger sweeps remain useful as empirical context: they suggest that the formal five-witness family is not isolated and that non-vacuous PAC-Bayes-kl behavior persists across a much broader range of modern models. However, the theorem-level route to resolving the three criteria in this paper is intentionally restricted to the explicitly certified witness families $\Lambda_{\mathrm{Img}}^{(5)}$ and $\Lambda_{\mathrm{Img}}^{(4)}$.

Interpretation. The formal witness family is the intended existential witness for the closure question posed in Section 1: with the audit-side certification step above discharged by the audit run, there exists a single PAC-Bayes framework and an explicit finite ImageNet-scale family for which we simultaneously obtain formal non-vacuity, polynomial-time certificate evaluation, and audit-certified predictive tightness. The broader MNIST/ImageNet sweeps are read as supplementary empirical evidence rather than as the core proof witness.

Full experimental details, figures, and pairwise concordance analysis are in the companion paper [3].

Remark 12.2 (Formal non-vacuity witness with tracked error tolerances)**.** *The plug-in convention $\varepsilon_{mc} = 0$, $\eta_{num} = 0$ follows standard practice in the PAC-Bayes empirical literature [2, 7]. The table above exhibits five formal witnesses spanning all four architecture families; here we detail the convergence behavior for EfficientNet-B0 as a representative example. At $\sigma = 0.001$ (no fine-tuning), with $m = 10$, $\delta_{pb,\lambda} = 1.5625 \times 10^{-3}$ (Bonferroni allocation over Λ_{80}), $\delta_{mc} = 0.025$, $\eta_{num} = 10^{-10}$: Hoeffding gives $\varepsilon_{mc} = \sqrt{\ln(2/\delta_{mc})/(2m)} \approx 0.468$, yielding $p_{\mathrm{cert}} = \hat{L}_{S_1}^{(m)}(Q) + \varepsilon_{mc} \approx 0.693$. With $\mathrm{KL} = 0$ and $C(n_1, \delta_{pb,\lambda}) \approx 11.96$, the PAC-Bayes slack is $\varepsilon_{pb}^{\mathrm{cert}} \approx 7.97 \times 10^{-4}$, giving $B_{\mathrm{ana}} = \mathrm{kl}_+^{-1}(p_{\mathrm{cert}}, \varepsilon_{pb}^{\mathrm{cert}}) \approx 0.711$ and returned endpoint $B_{\mathrm{cert}} \in [B_{\mathrm{ana}}, B_{\mathrm{ana}} + 10^{-10}] \approx 0.711 < 1$. The formal certificate is looser than the plug-in value (0.711 vs 0.239) because ε_{mc} dominates with only $m = 10$ samples, but it remains strictly non-vacuous. Increasing m to 100 yields $\varepsilon_{mc} \approx 0.148$ and $B_{\mathrm{cert}} \approx 0.392$; to 1,000, $\varepsilon_{mc} \approx 0.047$ and $B_{\mathrm{cert}} \approx 0.290$, converging toward the plug-in value of 0.239.*

Remark 12.3 (Legacy statistical certification on the no-fine-tune 40-configuration subset)**.** *An earlier draft of this work used a broader statistical certification argument on the 40-configuration no-fine-tune subset. We retain it here as supplementary context rather than as the primary theorem-level criterion 3 witness. The key conclusion was that the formal certificate ranking on that subset remains highly aligned with the true-risk ordering, with formal Kendall values around* 0.97 *under the corresponding finite-m allocation.*

13 Conclusion

We have established a rigorous sample-split PAC-Bayes theorem package in the Gibbs-predictor setting and discharged its audit-side closure on an explicit ImageNet family. The ingredients are:

1. the PAC-Bayes-kl inequality with Maurer's constant,
2. certified kl-inverse computation with explicit Monte Carlo and numerical error tracking,
3. polynomial-time certificate evaluation under explicit sampling, KL-computability, and certified-arithmetic assumptions,
4. and an independent-audit theorem that converts simultaneous audit intervals into a rigorous lower bound on Kendall's τ_a between certificate values and true risks.

At theorem level, the explicit ImageNet witness family

$$\begin{aligned}\Lambda^{(5)}_{\mathrm{Img}} = \{&\text{ConvNeXt-Large},\ \text{ViT-L/16},\ \text{Swin-Tiny},\\ &\text{EfficientNet-B0},\ \text{ResNet-18}\} \times \{\sigma = 0.001,\ \text{no fine-tuning}\}\end{aligned}$$

provides five fully tracked returned certified endpoints, all strictly below 1:

$$0.622,\ 0.632,\ 0.675,\ 0.711,\ 0.778.$$

Using an independent audit set of 15,000 images, the audit run instantiates the Monte Carlo regime via the two-sided empirical-Bernstein concentration of Audibert–Munos–Szepesvári (2009, Theorem 1) with pre-declared per-witness draw counts and produces certified eB-padded audit intervals that directly separate 9 of the 10 unordered pairs (the only overlapping pair is {ConvNeXt-Large, ViT-L/16}, anticipated). This discharges the lower bound

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}}) \geq 0.8.$$

On the same audit event, the deterministic-restriction corollary to the four-configuration subset $\Lambda^{(4)}_{\mathrm{Img}}$ (Section 12) further yields perfect certified rank agreement,

$$\tau_a(B_{\mathrm{cert}}, L_{\mathcal{D}})\big|_{\Lambda^{(4)}_{\mathrm{Img}}} = 1,$$

as a post-hoc observation without additional probability cost.

The theorem chain therefore isolates a concrete sample-split Gibbs theorem package for the open problem, and the explicit ImageNet witness family with its discharged audit certificate satisfies the three target criteria in a fully theorem-level form:

- formal non-vacuity,
- polynomial-time certificate evaluation,
- and certified predictive tightness.

Larger MNIST and ImageNet sweeps, including 90/90 non-vacuous plug-in values, are also reported as supplementary empirical evidence. Those broader experiments suggest that the formal witness family is representative of a much wider phenomenon; the theorem-level claim of this paper is deliberately confined to the explicitly certified finite families above with their discharged audit certificate.

A stronger future program would replace the present sample-split witness by a more standard full-data PAC-Bayes regime and add an explicit transfer theorem from Gibbs risk to the corresponding deterministic predictor. This is separate from the closed theorem package of the present proof note.

A Witness manifest and audit-side closure artifact

The explicit theorem-level witness used in Section 12 is the no-fine-tuning ImageNet family

$$\Lambda_{\mathrm{Img}}^{(5)} = \{\text{ConvNeXt-Large}, \text{ViT-L/16}, \text{Swin-Tiny}, \text{EfficientNet-B0}, \text{ResNet-18}\} \times \{\sigma = 0.001\}.$$

The deterministic-restriction corollary uses the four-element subset

$$\Lambda_{\mathrm{Img}}^{(4)} = \{\text{ConvNeXt-Large}, \text{Swin-Tiny}, \text{EfficientNet-B0}, \text{ResNet-18}\} \times \{\sigma = 0.001\},$$

which is not a separate pre-registered witness family but a post-hoc restriction on the $\Lambda_{\mathrm{Img}}^{(5)}$ audit event.

Concrete protocol. The current witness construction uses split seed 123 with

$$|S_0| = 20{,}000, \qquad |S_1| = 15{,}000, \qquad |T| = 15{,}000.$$

For each architecture λ, the prior/posterior pair is

$$P_\lambda = Q_\lambda = \mathcal{N}(w_\lambda^{\mathrm{ref}}, \sigma^2 I), \qquad \sigma = 0.001,$$

so no fine-tuning is used and $\mathrm{KL}(Q_\lambda \| P_\lambda) = 0$.

Certificate-side parameters. The displayed returned certified endpoints use

$$m = 10, \qquad \delta_{pb,\lambda} = \tfrac{0.125}{80} \approx 1.5625 \times 10^{-3}, \qquad \delta_{mc} = 0.025, \qquad \eta_{num} = 10^{-10}, \qquad \alpha = 0.05,$$

where $\delta_{pb,\lambda}$ is the per-configuration PAC-Bayes budget allocated uniformly across the full 80-configuration sweep Λ_{80} via the Bonferroni union bound of Section 12 (S_1-side hardening), δ_{mc} is the per-displayed-configuration Monte Carlo budget, and α is the familywise audit failure budget. For the audit intervals in the exact-loss regime,

$$t = \sqrt{\frac{\ln(200)}{30{,}000}} \approx 0.01329.$$

Audit-run summary. The audit-side closure is discharged by an audit run on the ImageNet-1K validation split using $3 \times$ NVIDIA H100 80GB HBM3, total walltime 38.24h. The pre-declared per-witness draw counts $m_a(\lambda) \in \{1{,}625, 2{,}380, 8{,}400, 7{,}820, 1{,}400\}$ are fixed before any audit observation. For each $\lambda \in \Lambda_{\mathrm{Img}}^{(5)}$, the realized unrounded audit-MC mean $\hat{L}_T^{(m_a)}(Q_\lambda)$, sample variance $s_a^2(\lambda)$, empirical-Bernstein half-width $\varepsilon_{\mathrm{audit,mc}}^{\mathrm{eB}}(\lambda)$, and padded audit interval I_λ are displayed in the table of Section 12, together with the per-witness Hoeffding half-width $t_H = 0.014132$ and the bottleneck pairwise separation 1.10×10^{-3}. The certified pair count is 9/10 on the unrounded eB-padded intervals (only {ConvNeXt-Large, ViT-L/16} overlaps); restricting to the four-element subfamily $\Lambda_{\mathrm{Img}}^{(4)}$ gives all $\binom{4}{2} = 6$ pairs separated.

Reproducibility envelope. The dataset is the ImageNet-1K validation set (50,000 images), partitioned by seed 123 into $S_0/S_1/T$ of sizes 20,000/15,000/15,000. Pretrained reference weights w_λ^{ref} are publicly distributed checkpoints for each architecture, trained exclusively on the disjoint ImageNet-1K training split (approximately 1.28M images), with no validation-set leakage and no distillation from validation-trained models; we treat each w_λ^{ref} as a deterministic constant chosen prior to drawing (S_0, S_1, T), so the prior $P_\lambda = \mathcal{N}(w_\lambda^{\mathrm{ref}}, \sigma^2 I)$ is data-independent in the formal sense required by Theorem 6.1. A single master base seed $b_0 = 123$ is logged in the run artifact `final_audit.json`. The artifact records b_0 together with the realized per-shard audit seeds (e.g., `2606300848`, `42401006`, `1314453031` for ConvNeXt-Large shards $0, 1, 2$) and the realized certificate-side draw transcripts. The mathematical theorem is stated for ideal independent random variables $U_{\mathrm{plug-in}}$, U_{cert}, U_{audit} as in Section 3; the logged SHA-256 streams are an implementation of the ideal-randomness assumption used to reproduce the realized experiment bit-for-bit, not a measure-theoretic proof of independence. In the simulation code the three streams are realized by deterministic SHA-256-truncated hashing of b_0 with disjoint per-stream tags; bit-exact replay is supported by the master seed, the per-shard seed log, and the published source.

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